1 Approximate laws and the best system

Kepler's laws are false. Against a modern planetary theory, a fixed ellipse misses Mars by one or two arcminutes over a few decades and the Moon by about a degree within a month (**??**); yet the laws were, and are, good science. A proof-requiring inducer seems ill suited to such laws: a deterministic theory either proves a datum or does not, and an exact law that is slightly wrong is refuted by any datum precise enough to show its error. This section asks what SAI does with an approximately true law, and what becomes, inside it, of Lewis's best system and of explanatory power.[1] In brief, the difficulty is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view as such: M_A mixes randomised theories, and under SAI's pricing an approximately true law is a law with a calibrated error model (Proposition 1.3). These results carry the row on approximate truth of the argument map (**??**). Appendix A states in full the results that this section states in words, and proves all of them.

1.1 Setting

The data are answers to questions (**??**), such as the cell of width Δ in which the longitude of Mars lies at time t (**??**).

Definition 1.1 (Answered data, exception codes, approximate laws)**.**

(a) *Answered data.* A datum is a pair $(\mathcal{Q}, a)$ of a question and its true answer; a data set D has pairwise distinct questions and is read as the two-valued data set $\{(a, 1) : (\mathcal{Q}, a) \in D\}$; $\operatorname{dom} D$ lists its questions, y_D their answers, and $n := |D|$. A theory T *proves* (*refutes*) the datum if $B \cup T \vdash a$ ($B \cup T \vdash \neg a$), and is *silent* otherwise; $A_T(\mathcal{Q}) := \{a' \in \mathcal{Q} : B \cup T \nvdash \neg a'\}$, and $\gamma(\mathcal{Q}) := \log|\mathcal{Q}|$ is the *literal cost*.

(b) *Codes.* For a prefix-free code c_S of sentences, $\ell(\chi) := 1 + |c_S(\chi)|$, and $\|\Gamma\| := \sum_{\chi\in\Gamma} \ell(\chi)$ for a finite set Γ of sentences (a theory or a *module* of one). If $B \cup T \cup D^{\pm}$ is consistent, the *exceptions* of T are the data it does not prove, priced by $\mathrm{Exc}^{Q}_{D}(T) := \sum \gamma(\mathcal{Q})$ (code Q), $\mathrm{Exc}^{F}_{D}(T) := \sum \ell(a)$ (code F) or $\mathrm{Exc}^{R}_{D}(T) := -\log P^{R}_{T}(D)$ (code R, below); otherwise $\mathrm{Exc}_D(T) := \infty$. Unmarked, Exc is code Q. $J_\lambda(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T)$ for $\lambda > 0$ (and J^{Q}_{λ}, J^{F}_{λ}, J^{R}_{λ}); its minimisers are the *λ-MAPs* (Hänni and Claude (Anthropic), 2026, Definition 3.6). T^{-Z} withdraws T at the finitely many times in Z, at about c_{time} bits per time.

(c) *Interval laws.* $B :=$ RCF (decidable and complete; standard, Tarski, 1951), with constants q_t for times t in a decidable set I of numerals, and a rational range $[0, \mathsf{R}]$. A *pointwise interval law* is $\{\underline{f}(t) \le q_t \le \overline{f}(t) : t \in I\}$ for total computable $\underline{f} \le \overline{f} : I \to \mathbb{Q} \cap [0, \mathsf{R}]$ named by a codeword of a fixed decidable prefix-free presentation language; $\|T\|$ is the codeword's length. The *exact law* T_0 has $\underline{f} = \overline{f} = f$, the *honest law* T^{ε} has $f \mp \varepsilon$ (clipped). *Interval data* answer questions that partition $[0, \mathsf{R})$ into finitely many cells with rational endpoints; with C_t the intersection of the cells that D assigns to t, $P^{R}_{T}(D) := \prod_t |C_t \cap [\underline{f}(t), \overline{f}(t)]|/(\overline{f}(t) - \underline{f}(t))$ (length; the factor is $\mathbf{1}[\underline{f}(t) \in C_t]$ if $\underline{f}(t) = \overline{f}(t)$).

(d) *Errors.* For a real quantity $q(t)$, $e(t) := f(t) - q(t)$, and $\varepsilon^* := \operatorname{ess\,sup}|e|$ under the law of the error at a data time. *Regime G:* datum j is "$q(t_j) \in [v_j - \delta, v_j + \delta]$" with $v_j = q(t_j) + \zeta_j$ rational and $|\zeta_j| \le \delta_0 \le \delta$. *Regime R: q* lives on a circle of circumference R; datum j is about a *fresh*

[1] The questions of approximately true laws and of the domain of applicability of a model are discussed in Hänni's working notes (Hänni, 2026).

time t_j (an injective computable sequence of numerals); the errors $e_j := e(t_j)$ are i.i.d.; and datum j is the cell of width Δ (R/Δ an integer) containing $q(t_j)$ in a grid shifted by a dither u_j, uniform and independent of everything else; $\gamma = \log(\mathsf{R}/\Delta)$, $p_\Delta := \mathbb{E}\min(|e|/\Delta, 1)$ and $s_\Delta := \min(2\varepsilon^*/\Delta, 1)$.

(e) *Peakedness.* $\mathrm{pk}(e) := \varepsilon^*/\mathbb{E}|e|$ and $\mathrm{pk}_\varepsilon(e) := \varepsilon/\mathbb{E}|e|$; every reported value names its fit (minimax or mean).

1.2 Deterministic theories and precision

Over deterministic theories, proof-requiring compatibility is monotone in logical strength, so data never move the odds towards a weaker theory: $T_0 \vdash T^\varepsilon$, every data set compatible with T^ε is compatible with T_0, and with prior weights $2^{-\|T\|}$ the posterior odds of T_0 against T^ε are at least 2 on each, although T_0 is false when $e \not\equiv 0$ (Proposition A.9). Under guaranteed tolerances (regime G), T_0 is almost surely compatible with the data at every sample size when $\delta \geq \varepsilon^* + \delta_0$ and, under mild conditions on the measurement error, is eventually refuted when $\delta < \varepsilon^* + \delta_0$ (Proposition A.11). Rounding changes the picture.

Proposition 1.2 (Dithered rounding)**.** *In regime R, given $e(t_j)$, T_0 proves datum j with probability $(1 - |e(t_j)|/\Delta)^+$ and refutes it otherwise. So T_0 survives n data with probability $(1 - p_\Delta)^n$, its first refutation has a geometric index with mean $1/p_\Delta$, and it is almost surely refuted eventually iff $\mathbb{P}[e \neq 0] > 0$, for every Δ, however coarse. For $\Delta \geq \varepsilon^*$, $p_\Delta = \mathbb{E}|e|/\Delta$: the exact law survives about $\Delta/\mathbb{E}|e|$ data.*

This corrects the natural expectation that coarse data protect a false exact law: precision belongs to the data set, not to the datum, and an exact law with a sinusoidal error of amplitude $1'$ is refuted by positions rounded to a dithered grid of 1° after about 94 data (the proof and the comparison of several exact laws, Proposition A.12, are in Appendix A.3). We have not found these two regimes stated elsewhere; precision-dependent selection as such is familiar from accounts based on AIC, MDL and MML (cf. Forster and Sober, 1994; Rissanen, 1978; Wallace, 2005).

1.3 Randomised theories and laws with error models

A deterministic theory pays, in code F, for which questions are exceptions (Proposition A.2). A theory machine that reads a random tape answers every question at once, at random inside what the law allows: for a pointwise interval law T, $L_A(D) \leq \|T\| + \mathrm{Exc}_D^R(T) + c$ on interval data consistent with T (Proposition A.3). In fact $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$ over theory machines G: SAI is the best two-part code over randomised theories, neither K_A nor literal-exception codes over finite theories are within an additive constant of it in general, and over all theory machines a steeper penalty collapses onto the universal machine, so best-system analyses use laws with private literal facts (Propositions A.4 and A.7). *SAI's pricing* of a form is the weight $2^{-(K(G) - \log \nu_G(D))}$ of a named theory machine G in M_A's mixture, not the whole posterior.

With exceptions named in the law at c_{time} bits each (a cost model valid only while times are that cheap to name; Proposition A.15), the honest law T^{ε^*} has a lower rate than the patched exact law T_0^{-Z}, for $\Delta \geq 2\varepsilon^*$, iff $\lambda c_{\mathrm{time}} > \gamma(2\mathrm{pk}(e) - 1)$: spread errors favour honesty and peaked ones exceptions (Propositions A.16 and A.17). Under random completion silence costs at most $\log \mathrm{e}$ bits per straddled cell boundary, e Euler's number (Lemma A.18), and the comparison moves inside the band. For $\varepsilon \geq \varepsilon^*$ and a computable distribution $\mathcal{G}$ on $[0, 1]$, the *band-confined completion*

$\mathrm{BC}_{\mathcal{G}}$ emits T^{ε} and, for each time t_i, the value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$ with $\hat{w}_i \sim \mathcal{G}$ drawn from the tape. On data at fresh times its law is $\prod_j \mathcal{G}(C'_j)$, with C'_j the cell of datum j in band coordinates (the band mapped affinely onto $[0,1]$), and $K(\mathrm{BC}_{\mathcal{G}}) \le \|T^{\varepsilon}\| + K(\mathcal{G}) + c$. Uniform $\mathcal{G}$ gives the *uniform band*; $\mathrm{HY}(\rho)$, with $\mathcal{G} = (1-\rho)\delta_{1/2} + \rho\,\mathrm{Unif}[0,1]$, is "exact, except in a share ρ of cases, and then within ε".

Proposition 1.3 (Laws with error models)**.** *In regime R let $\varepsilon \ge \varepsilon^*$, and let $\mathcal{F}$ be the distribution function of $w_j := (\varepsilon - e_j)/(2\varepsilon)$, the position of the true value in the band.*

(a) If $\Delta \ge 2\varepsilon$, the rate of $\mathrm{BC}_{\mathcal{G}}$ is $(2\varepsilon/\Delta)\, c_{\mathcal{G}}$, with

$$c_{\mathcal{G}} := \mathbb{E}_w \int_0^1 \left(\mathbf{1}[v < w](-\log(1 - \mathcal{G}(v))) + \mathbf{1}[v > w](-\log \mathcal{G}(v))\right) dv.$$

So band-confined completions are ranked in the same way at every such Δ.

(b) $c_{\mathcal{G}} \ge c_{\mathcal{F}} := \int_0^1 H_2(\mathcal{F}(v))\, dv$, H_2 the binary entropy, with equality iff $\mathcal{G} = \mathcal{F}$ almost everywhere on $[0,1]$. In particular the uniform band is the best band-confined completion iff e is uniform on $[-\varepsilon, \varepsilon]$.

(c) For every theory machine G', G_U included, almost surely $\liminf_n \frac{1}{n}(-\log \nu_{G'}(D_n)) \ge r_{\mathcal{F}}(\Delta) := \mathbb{E}[-\log \mathcal{F}(C'_1)]$, the conditional entropy rate of the answers given the questions, while $-\frac{1}{n}\log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n) \to r_{\mathcal{F}}(\Delta)$. If $\mathcal{F}$ is computable, then $L_A(D_n)/n \to r_{\mathcal{F}}(\Delta)$ almost surely. For $\Delta \ge 2\varepsilon$, $r_{\mathcal{F}}(\Delta) = (2\varepsilon/\Delta)\, c_{\mathcal{F}}$.

(d) $c_{\mathrm{HY}(\rho)}$ is convex in $\rho \in (0,1]$, and some $\mathrm{HY}(\rho)$ with $\rho < 1$ beats the uniform band iff $\mathbb{E}\ln(1 + |e|/\varepsilon) < 2\ln 2 - 1$. This requires $\mathrm{pk}_{\varepsilon}(e) > \ln 2/(2\ln 2 - 1) \approx 1.79$, and holds whenever $\mathrm{pk}_{\varepsilon}(e) > \mathrm{e}/(4 - \mathrm{e}) \approx 2.12$.

Proof idea. A cell boundary falls into the band with probability $2\varepsilon/\Delta$, at a uniform position v, and the datum then costs $-\log$ of the $\mathcal{G}$-mass on the true side; (b) is Gibbs' inequality given v; (c) uses that a theory proves at most one answer list, with Markov's inequality and the Borel–Cantelli lemma; (d) is convexity and the sign of the derivative at the uniform band (Appendix A.4).

So under SAI's pricing an approximately true law is a law with a calibrated error model, not a silent partial theory and not an exact law with listed exceptions; the dichotomy between the two survives inside the band, with peakedness thresholds near 2. That a model should include its noise distribution is the standard situation of MDL (cf. Rissanen, 1978; Grünwald, 2007); what is specific is the reading for a proof-requiring inducer, in which the honest partial theory is one completion among many. Selection among finitely many forms is almost sure (Theorem A.20 and Corollary A.21). For deterministic theories the trichotomy is the formal home of humility: on a finer question, the honest and the empty law put their posterior mass on "independent", and whether the inducer ends humble is decided not by the prior, which favours the exact law, but by the error's shape and the price of silence (Proposition A.22). For the randomised theories that M_A mixes, humility is calibration: by (c) the preferred spread over the cells is the true error distribution.

Kepler's residuals. For seven bodies and six windows each (three for the Moon) we kept the fixed Kepler ellipse with the least largest longitude error (the minimax fit) and the one with the least mean error (the mean fit), and took their errors at 8000 uniform times as error laws, against the plan94 and moon98 ephemerides (Table 1 in Appendix A.4). The errors are moderately spread: the peakedness is 1.60–3.95 for the minimax fit and 1.89–7.08 for the mean fit, against

$\pi/2$ for a sinusoid and 2 for uniform $|e|$. A computation of the rates at eleven cell widths from $0.1'$ to 30° (code in the accompanying repository; details in Appendix A.4) gives the following. Among the Kepler-based theory machines compared, the law with a 40-bin error histogram, fitted on half of the times and scored on the other half, beats the uniform band and the exact law with exceptions in all 429 (body, window, cell width) cells, by 0.4% to 25.7% of the band's rate, although the uniform band wins every cell within the pool $\{\emptyset, T_0^{-Z}, \text{uniform band}\}$; and in the coarse regime the two-point form beats the band for 34 of the 39 body–window pairs, exactly where Proposition 1.3(d) says. The residuals are deterministic functions of time, so a better theory (Newton's, or the ephemeris itself) eventually beats every Kepler-based machine.

1.4 Sound reasoning with approximate models

If the background is true, true honest models of one quantity are jointly consistent, and every statement about the quantity derived from them is a set containing its true value. If the background also decides the comparisons involved (as RCF does for semialgebraic models), exact approximations that differ anywhere on a common domain are jointly inconsistent and so prove everything, while joint consistency of their honest forms is a pairwise check of tolerances in one dimension and for boxes, and an $(m+1)$-wise check for Euclidean balls in $\mathbb{R}^m$, by Helly's theorem (Proposition A.25). An honest dynamical law with initial tolerance $\varepsilon^{(0)}$ and tolerance ε per step proves, after r steps of an L-Lipschitz map, the tolerance $L^r\varepsilon^{(0)} + \varepsilon(L^r - 1)/(L - 1)$ $(L \neq 1)$, tightly for linear maps in one dimension: it has a derivation horizon, and an exact approximation is guaranteed correct up to that horizon (for linear maps in one dimension, only up to it) (Propositions A.23 and A.24). The device is sound, but it does not say how to choose a model, a domain or a tolerance (**??**).

Remark 1.4 (Typical against worst case). With $L = 1$ and $r = 100$ steps the propagated tolerance is 100ε, while with independent random-sign errors $\pm\varepsilon$ the mean deviation is $\varepsilon r\binom{r}{r/2}2^{-r} \approx 7.96\varepsilon$: a probabilistic error model answers much further out, with a stated confidence instead of a guarantee.

1.5 The best system as a steeper MAP

On Lewis's account the laws are the theorems of the deductive system that best balances simplicity against strength (cf. Lewis, 1983, 1994). Take strength to be fit to data in a fixed base language, the balance to be λ, and the candidates a class of laws with private literal facts; the relation to Lewis is an analogy, since best systems are a metaphysics of laws and SAI is inference from finite data. The points $(\|T\|, \mathrm{Exc}_D(T))$ have a convex hull with vertices v_i in order of length, joined by edges of slopes $-\sigma_i$, $\infty =: \sigma_0 > \sigma_1 > \cdots > 0$; for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are the theories at v_i (Hänni and Claude (Anthropic), 2026, Lemma 3.8). A point $(\alpha_\mathrm{r}, L_\mathrm{r})$ attained by some theory is a *zero-window* (σ, τ)*-kink* (Hänni and Claude (Anthropic), 2026, Definition 5.2) if every theory shorter by $x > 0$ has fit at least $L_\mathrm{r} + \sigma x$ and every theory longer by $x > 0$ has fit at least $L_\mathrm{r} - \tau x$.

Proposition 1.5 (Kind nature is a kink)**.** *For a family of candidate theories satisfying the hypotheses of the hull lemma (Proposition A.7) and $0 \le \tau < \sigma \le \infty$, the following are equivalent: (a) the set of λ-MAPs is the same for every $\lambda \in (\tau, \sigma)$; (b) some hull vertex v_i has $\sigma_{i+1} \le \tau$ and $\sigma \le \sigma_i$; (c) the point family has a zero-window (σ, τ)-kink at a point that is a hull vertex.*

Proof idea. The λ-MAP set is fixed between consecutive edge slopes and changes at each; (b)⇔(c) is Hänni and Claude (Anthropic) (2026, Corollary 5.4(c)) (Appendix A.6).

So nature is kind on an interval of balances exactly when the data's hull has a kink there. In a solar-system toy with eight planets of 655 to 3838 observations, sixty asteroids with a median of about five and thirty comets with one or two (Proposition A.29), the λ-MAP is "the law plus the elements of all eight planets" for every $\lambda \in (7.03, 135.37)$, while below $\lambda \approx 7$ it changes with every asteroid (code in the accompanying repository). Inside such an interval the worry that the best system depends on an arbitrary choice of balance (cf. Woodward, 2014) is moot, and kindness is a checkable property of a language, a family of candidates and a data set. Lewis's trivialisation worry, that a predicate true of exactly the actual things makes its universal closure maximally simple and strong (cf. Lewis, 1983), is blocked by the coding bound, up to the complexity of a language chosen after the data (Proposition A.27).

Definition 1.6 (Explanatory power)**.** For a theory T and a module $\mathcal{A} \subseteq T$, $\mathrm{EP}_D(\mathcal{A}; T) := (\mathrm{Exc}_D(T \setminus \mathcal{A}) - \mathrm{Exc}_D(T))/\|\mathcal{A}\|$: the bits of data the module explains per bit of its statement, in the context T.

Every module of a λ-MAP has EP $\geq \lambda$, no addition to it has EP $> \lambda$, and for $\lambda > 1$ a fact used once never enters (Propositions A.26 and A.28). So the model–randomness split of the companion paper separates facts reused across many data from facts used once, not laws from initial conditions: in the toy, the elements of a well-observed body explain hundreds of bits per bit and sit on the steep side of the hull with the law (Proposition A.29). What singles out the law is how its EP scales: with m_{b} bodies of n_0 observations each, all stated, a law of L bits has $\mathrm{EP} = m_{\mathrm{b}} n_0 \gamma / L$, which grows with the number of systems, while each body's elements explain only that body's observations. Kepler's third law K3 is a law about initial conditions: it shortens every elements module by s bits, and when the set S of bodies whose elements are stated is the same with and without it, it lowers the objective iff $s|S|$ exceeds its length, whatever λ is; this compares law choices at exchange rate 1 and is not membership of the λ-MAP. In the question format, elements are question antecedents and never enter a λ-MAP (Proposition A.30). **??** computes what K3 is worth in Kepler's setting.

1.6 Explanation and correction

$\|\mathcal{A}\| \, \mathrm{EP}_D(\mathcal{A}; T)$ is the log-likelihood ratio of T against $T \setminus \mathcal{A}$ for $P_T(D) := 2^{-\mathrm{Exc}_D(T)}$. The support a hypothesis gets from joint data $D_1 \cup D_2$ splits into the supports from the parts plus a term U, the change in the mutual information between D_1 and D_2 that the hypothesis brings about, which has the form of Myrvold's measure of unification (Myrvold, 2003): with stated parameters, for laws that are products over questions, $U = 0$, and when a k-bit parameter is integrated out, U is k within one system, 0 across systems with independent parameters, and the bits of a shared constant across systems that share one (Proposition A.31). At $\lambda = 1$ a module pays iff its EP exceeds 1, the compression criterion of inductive logic programming (cf. Muggleton, 1995; Srinivasan et al., 1994). Copernicus against Ptolemy is unification without better prediction: against Ptolemy with the coincidence of each superior planet's retrograde motion and opposition stated as an axiom, at a_p bits each, which predicts the same, Copernicus, at H extra law bits, wins at every λ iff $H < \sum_p a_p$, and against Ptolemy with free phases it wins iff $\lambda < \sum_p a_p / H$ (Proposition A.32; the lengths are not measured; for the history, see Kuhn, 1957).

Newton's theory does not entail Kepler's laws; it contradicts them (cf. Duhem, 1954) and derives corrected statements, as in reduction to a corrected reduced theory (cf. Schaffner, 1967). In an exact rational toy, uniform circular motion perturbed to $x_{t+1} = Ax_t$ with A a rotation plus $\eta_{\mathrm{c}}\mathbf{P}$, the successor proves the honest old law on the window $t \leq \bar{t}$ with uniform tolerance ε iff

$\varepsilon \geq \varepsilon_{\min}(\bar{t})$, where $\varepsilon_{\min}(\bar{t}) \leq ((1+\eta_c\|\mathbf{P}\|)^{\bar{t}}-1)|a|$ for the initial state a, and, if $\eta_c\mathbf{P}a \neq 0$, it refutes the exact old law; among uniform-tolerance forms, the strongest it proves is the strongest both theories prove, while time-dependent tolerances are stronger common consequences and the whole common content is the disjunction of the two theories (Proposition A.33). The old law's domain of applicability is derived, not stipulated. Deriving "Kepler within ε on a window" from Newton needs validated perturbation bounds (**??**).

1.7 Domains of applicability and retention

A domain restriction that leaves one system's data literal pays asymptotically iff the law, even with an honest tolerance for that system, is worse than literal data, exact ties being decided by the constants (Proposition A.35). Under SAI's pricing that never happens, at cell widths below R/e, for a system that the law confines within about 66° of longitude (a sufficient bound): an honest band of half-width ε^* costs at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$ bits per datum, below the literal $\log(\mathsf{R}/\Delta)$ iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$. So the Moon ($\varepsilon^* = 2.6^\circ$ over a year) is kept in Kepler's law with its own tolerance at every cell width computed, from $10'$ to 30°, whereas under literal pricing at $10'$ and $60'$ a cheap domain clause excludes it. A superseded theory is retained where its derivation savings exceed its loss of fit, for all cell widths above a computable Δ^* within the range where its rate is monotone (Proposition A.36).

Domains are learned as tolerances. We conclude that under SAI's pricing a domain of applicability shows up as a tolerance and an error distribution that depend on the system and the window, not as a sharp boundary: for every planet and window computed, and for the Moon over a month, a year and 18.6 years, a law with its own tolerance has a lower rate than leaving the body's data literal. A sharp boundary is selected only when even an honest tolerance is worse than literal data, or under literal pricing at fine precision. The posterior over an explicit pool of domain predicates is not computed (Question A.37).

A Proofs for approximate laws and the best system

This appendix states in full the results of Section 1 that the main text states in words, and proves every result of that section. The notation is that of Definition 1.1. We measure a finite theory by $\|T\|$, without the stop bit of a prefix-free code of theories; using the prefix-free length $\|T\|+1$ instead changes every statement below by an additive constant at most, and shifts J_λ by the same amount λ on every theory, so that it leaves the λ-MAPs unchanged. Computed numbers come from seeded scripts in the accompanying repository; they are never part of a statement.

A.1 The setting in full

Technical points. (i) *Rational cells.* In regime R the dither u_j is uniform on a grid $\{k/N_\mathrm{d} : 0 \leq k < N_\mathrm{d}\}$, so that cell endpoints are rational. Every event used below has the form "u_j lies in an interval of length ℓ'", whose grid probability is within $1/N_\mathrm{d}$ of ℓ'; the formulas are the limit $N_\mathrm{d} \to \infty$ (equivalently, endpoints may be real algebraic numbers). A cell that crosses the zero point of the circle is a disjunction of two interval sentences. (ii) *Fresh times.* Data are sentences, so they mention numerals, a countable set; no atomless law lives on it. Times drawn i.i.d. from a law on numerals repeat: the number of distinct times among n draws is $o(n)$ almost surely (take a finite set of mass at least $1-\epsilon$ and apply the strong law of large numbers). A repeat changes

the cost of every form except an exact law: a patched law withdraws the law at a whole time, so every later datum at a refuted time is an exception, and the cells of an honest law at one time intersect, which narrows its band. A computation with times drawn uniformly from the 4383 days of a twelve-year window (sinusoidal error, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$, $c_{\text{time}} = \log 4383 + 2$) gives a patched law's cost per datum of 0.40, 1.96 and 3.23 bits at $n = 10^3$, 10^5 and 2×10^5, against the rate 0.240 at fresh times, and a code-R cost of the honest law falling from 0.037 to 0.022, against 0.030. This is why regime R takes fresh times: an injective computable sequence of numerals, with i.i.d. errors and dithers. Fresh times have names that grow, and Proposition A.15 says what that costs. (iii) *Presentations.* No decidable set of programs covers all total computable maps, so the class of pointwise interval laws depends on the presentation language (for example, primitive recursive definitions, or programs paired with an explicit clock); every statement holds for each fixed choice. (iv) *A model of the data.* In regime R the error at a fresh time is a draw from a fixed law; for the Kepler residuals this law is the law of $e(t)$ for t uniform in the window. Answered data with a question $\mathcal{Q}$ are read through **??**: for consistent $B \cup T$, T proves $(\mathcal{Q}, a)$ iff $A_T(\mathcal{Q}) = \{a\}$, since $T \vdash a$ refutes every other answer by incompatibility, and refuting every other answer proves a because B proves $\bigvee \mathcal{Q}$.

Lemma A.1 (Decidability)**.** *Let T be a pointwise interval law and C a cell, a finite union of intervals with rational endpoints. Then T proves the datum "$q_t \in C$" iff $[\underline{f}(t), \overline{f}(t)] \subseteq C$, and refutes it iff $[\underline{f}(t), \overline{f}(t)] \cap C = \emptyset$; and $B \cup T \cup D^{\pm}$ is consistent iff, for every time t of D, $[\underline{f}(t), \overline{f}(t)]$ meets the intersection C_t of the cells that D assigns to t. All three are decidable from the presentation of T.*

Proof. Every choice of reals $q_t \in [\underline{f}(t), \overline{f}(t)]$ gives a model of $B \cup T$ in $\mathbb{R}$, and if the q_t also lie in the assigned cells, a model of $B \cup T \cup D^{\pm}$. This gives the "only if" of the first two claims (choose q_t outside the cell, respectively inside it) and the "if" of the third. Inclusion and disjointness of finite unions of rational intervals are Boolean combinations of inequalities between rational numerals, which RCF proves or refutes; this gives the "if" of the first two claims. If for some t the interval misses C_t, the finitely many inequalities involved are jointly refutable in RCF, which gives the "only if" of the third. The values $\underline{f}(t), \overline{f}(t)$ are computable from the presentation, so all three conditions are decidable. □

Proposition A.2 (Laws plus listed exceptions are deterministic hypotheses)**.**

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $L_A(D) - c \le K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$.

(b) For an r.e. T and $E \subseteq D$ with $B \cup T \cup E^{\pm}$ consistent and $B \cup T \cup E^{\pm} \vdash \psi_D$, $L_A(D) \le K_{\equiv}(T) + K(E) + c$.

Proof. (a) The finite set $T' := T \cup \{a : (\mathcal{Q}, a)$ a datum of D that T does not prove$\}$ is contained in the consistent set $T \cup D^{\pm}$, so $B \cup T'$ is consistent, and T' proves every datum. Concatenating the codewords of the sentences of T, each after a continuation bit, the codewords $c_S(a)$ of the exceptions, each after a continuation bit, and a stop bit gives a prefix-free code of T' of length $\|T\| + \mathrm{Exc}_D^F(T) + 1$, and a fixed program turns it into an index of T'. So $K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$. The instance form of dominance (**??**) gives $M_A(D) \ge 2^{-K(e)-c}$ for every index e of a theory consistent with B that proves ψ_D, hence $L_A(D) \le K_A(D) + c$. (b) is **??**(b). □

So code F pays for *which* questions are exceptions, since the answer sentence names its question; a deterministic theory cannot avoid this, because it does not see the questions.

A.2 Randomised theories and the best two-part code

Proposition A.3 (Randomised theories pay question-relative costs)**.** *Let T be a pointwise interval law and D a set of interval data consistent with T.*

(a) $L_A(D) \le K(T) + \mathrm{Exc}_D^R(T) + c \le \|T\| + \mathrm{Exc}_D^R(T) + c$.

(b) If D comes from a computable question family with one question $\mathcal{Q}_t$ per time, and $k_t(T) := |A_T(\mathcal{Q}_t)| \le |\mathcal{Q}_t|$, then $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \sum \log k_t(T) + c$, the sum over the data that T does not prove; in particular $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \mathrm{Exc}_D^Q(T) + c$.

(c) Code R charges an exception of a patched law, at a time where $\underline{f} = 0$ and $\overline{f} = \mathsf{R}$, exactly $\log(\mathsf{R}/|C_t|)$, which is the code-Q cost for equal cells; it charges the honest law $-\log(|C_t \cap [f(t) - \varepsilon, f(t) + \varepsilon]|/2\varepsilon)$, which is 0 when the band lies inside the cell.

Proof. (a) *The machine.* Enumerate I as $t_1, t_2, \ldots$ and fix a computable injection $(i, k) \mapsto \langle i, k\rangle$ into tape positions, increasing in k. On a tape ω put $U_i := \sum_{k \ge 1} \omega_{\langle i,k\rangle} 2^{-k}$; under $\mathbb{U}$ the U_i are independent and uniform on $[0, 1]$. Let G_T be the theory machine that emits the sentences of T at the empty prefix and, once the prefix read contains the positions $\langle i, 1\rangle, \ldots, \langle i, k\rangle$, whose bits form the binary numeral of an integer m, also emits

$$\underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, m 2^{-k} \;\le\; q_{t_i} \;\le\; \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, (m+1) 2^{-k}.$$

This set of (prefix, sentence) pairs is c.e., and an index of it is computable from the presentation of T, so $K(G_T) \le K(T) + c$.

The random theory is always consistent. Put $q_{t_i} := v_i := \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i) U_i$. Each emitted closed interval contains v_i, so these values satisfy every emitted sentence, and $\mathbb{R}$ with them is a model.

It proves a datum "$q_{t_i} \in C$" whenever v_i lies in the interior of C. The emitted intervals contain v_i and have lengths $(\overline{f} - \underline{f})(t_i) 2^{-k} \to 0$, so one of them lies in the interior of C, and RCF proves the inclusion. (If $\underline{f}(t_i) = \overline{f}(t_i)$, T alone decides the datum.) The times of D are distinct indices i, the U_i are independent, and endpoints have probability 0, so

$$\nu_{G_T}(D) \;\ge\; \mathbb{U}\{\omega : v_t \in \operatorname{int} C_t \text{ for every time } t \text{ of } D\} \;=\; P_T^R(D).$$

Dominance (**??**), $M_A \ge 2^{-K(\nu)} \nu$ for every theory-machine law ν, gives $M_A(D) \ge 2^{-K(G_T) - c} \nu_{G_T}(D)$, which is (a); and $K(T) \le \|T\| + c$.

(b) Let the machine use U_i to pick the $\lfloor k_{t_i}(T) U_i \rfloor$-th element a of $A_T(\mathcal{Q}_{t_i})$, which is computable by Lemma A.1, and emit a as soon as enough bits have been read (almost surely after finitely many). The cell a meets $[\underline{f}(t_i), \overline{f}(t_i)]$, so the random theory has a model in $\mathbb{R}$; it proves the true answer with probability $1/k_{t_i}(T)$, independently across times, and $k_t(T) = 1$ when T proves the datum. Since D is consistent with T, T refutes no datum. The machine's index is computable from T and the question family, and dominance gives the bound; the last claim follows from $\log k_t(T) \le \log |\mathcal{Q}_t| = \gamma(\mathcal{Q}_t)$. (c) is the definition of P_T^R. □

Proposition A.4 (SAI is the best two-part code over theory machines)**.**

(a) For every data set D, $L_A(D) = \min_G [K(G) - \log \nu_G(D)] \pm c$, the minimum over theory machines G; it is attained, up to c, by G_U.

(b) Restricting G gives upper bounds: $L_A(D) \le \min_T J_1^R(T; D) + c$ over pointwise interval laws, $L_A(D) \le \min_T J_1^F(T; D) + c$ over finite theories, and $L_A(D) \le K_A(D) + c$.

(c) *Neither* K_A *nor* $\min_T J_1^Q(T;\cdot)$ *over finite theories is within an additive constant of* L_A *in general; nor, therefore, is* $\min_T J_1^F(T;\cdot)$ *when* $\gamma \le \ell$.

(d) *Over all theory machines steeper selection collapses. Take the point family of programs* $p \in \operatorname{dom} V$ *for theory machines, with length* $|p|$ *and fit* $-\log \nu_{G_{V(p)}}(D)$*, and let* ℓ_U *be the length of a program for* G_U*. For* $\lambda > 1$*, every* λ*-MAP* p *has* $|p| \le \ell_U + \ell_U/(\lambda - 1)$.

Proof. (a) Dominance gives $L_A(D) \le K(G) - \log \nu_G(D) + c$ for every theory machine G, and $M_A = \nu_{G_U}$ gives $K(G_U) - \log \nu_{G_U}(D) = L_A(D) + O(1)$. (b) is Propositions A.2 and A.3 with $K_{\equiv}$ and K of a finite theory at most its length plus a constant.

(c) That $K_A - L_A$ is unbounded is the absence of a coding theorem (**??**). For $\min_T J_1^Q(T;\cdot)$, take the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ with $B = \emptyset$ and the yes/no data $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$, so that every datum has literal cost $\gamma = 1$. By **??**, applied to the r.e. theory $\{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$, $K_A(D_n) \le c$, hence $L_A(D_n) \le K_A(D_n) + c \le c'$. Let T be finite with $B \cup T \cup D_n^{\pm}$ consistent. If c_j does not occur in T and $T \vdash R(c_j)$, then $T \vdash \forall x\, R(x)$ by the theorem on constants, which contradicts consistency with $\exists x\, \neg R(x)$. So T proves at most the data $R(c_j)$ with c_j occurring in T, together with $\exists x\, \neg R(x)$. In a syntax code in which each symbol occurrence costs at least one bit, each constant occurring in T costs at least one bit of $\|T\|$, and each unproved datum one bit of Exc^Q; so $J_1^Q(T; D_n) \ge n$. Finally $\mathrm{Exc}^F \ge \mathrm{Exc}^Q$ when $\gamma \le \ell$, since then each exception costs at least as much in code F.

(d) Hänni and Claude (Anthropic) (2026, Proposition 5.7(ii)) states: if a point family contains a member π_U, of length ℓ_U, with fit $L(\pi_U) \le L(\pi) + \ell(\pi) + \kappa_0$ for every member π, then for every $\lambda > 1$ every λ-MAP has length at most $\ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$. The mixture identity of **??** gives $M_A(D) \ge \sum_{p \in \operatorname{dom} V} 2^{-|p|} \nu_{G_{V(p)}}(D) \ge 2^{-|p|} \nu_{G_{V(p)}}(D)$ for each p, so a program p_U for G_U has fit $-\log \nu_{G_U}(D) = L_A(D) \le -\log \nu_{G_{V(p)}}(D) + |p|$ for every p. This is the hypothesis with $\kappa_0 = 0$. □

Remark A.5 (Pointwise interval laws: the separation is open). For every set D of interval data, both $L_A(D)$ and $\min_T J_1^R(T; D)$ over pointwise interval laws are at least $K(y_D \mid \operatorname{dom} D) - c$. For L_A: the answer conjunctions of the questions in $\operatorname{dom} D$ form a finite set S of pairwise B-incompatible sentences, computable from $\operatorname{dom} D$, and $M_A(D) = M_A(\{(\psi_D, 1)\})$ (**??**); so the coding lower bound of **??** gives $L_A(D) \ge K(\psi_D \mid S) - c \ge K(y_D \mid \operatorname{dom} D) - c'$. For J_1^R: Proposition A.8(b). So neither bound separates the two. The natural candidate for a separation, data in which pairs of times share a random cell, has $L_A \approx n\gamma/2$; but it does not separate them either, because a presentation may store values: the program that stores the $n/2$ shared cells and returns each for both times of its pair has length $n\gamma/2 + O(\log n)$ and $\mathrm{Exc}^R = 0$, so $\min_T J_1^R \le n\gamma/2 + O(\log n)$ as well.

Question A.6. Fix a presentation language for pointwise interval laws. Is $\min_T J_1^R(T; D) - L_A(D)$ bounded over all sets D of interval data? (For K_A and for $\min_T J_1^Q$ over finite theories the answer is no, by Proposition A.4(c).)

Proposition A.7 (Which hypotheses of the companion paper on steeper priors hold here)**.**

(a) Pointwise interval laws in code R *(a fixed presentation language; one question family with finitely many cells). The codewords form a decidable prefix-free set;* P_T^R *is computable uniformly in* T*, is the law* ν_{G_T} *of a theory machine (a shared hypothesis in the sense of Hänni and Claude (Anthropic), 2026, Definition 3.4) and, on data at distinct times, a product hypothesis (Hänni and Claude (Anthropic), 2026, Definition 3.3); Kraft's inequality holds, lengths are integers, and finitely many laws have each length. So Lemmas A.1, 3.8, 3.9 and 5.3, Corollaries 5.4 and 5.5 and Theorem 5.6 of Hänni and Claude (Anthropic) (2026) apply, with* $J_\lambda = J_\lambda^R$.

(b) Finite theories over a decidable consistent B, in code Q, *with* $P_T(D) := 2^{-\mathrm{Exc}^Q_D(T)}\mathbf{1}[B \cup T \cup D^{\pm}$ *consistent*$]$. *P_T satisfies the semimeasure inequalities of Hänni and Claude (Anthropic) (2026, Definition 3.4), and it is a product over questions when consistency can be checked question by question (pointwise laws at distinct times). It is not lower semicomputable in general (for $B = \emptyset$ and a language with a binary relation symbol), so this class is not a hypothesis class of the companion paper. Its hull lemma (Lemma 3.8), kink lemma (Lemma 5.3), Corollary 5.4, Theorem 5.6 and Lemma A.1(i) use only the point family $(\|T\|, \mathrm{Exc}^Q_D(T))_T$ and apply. Its coding bound (Lemma 3.9) is replaced by Proposition A.8, with which Corollary 5.5 holds with offset $b + \kappa + n$ in general, and $b + \kappa$ when provability of the answers from the theories is decidable (κ the constant of Proposition A.8).*

Proof. Which hypotheses are used. Each listed property is checked in the statement: for (a), $P^R_T = \nu_{G_T}$ on interval data by the proof of Proposition A.3(a) (the random theory proves "$q_t \in C$" iff $v_t \in \operatorname{int} C$, up to null events), and the product form holds because the U_i are independent. That the hull, kink and converse results need only the point family can be read off their statements, which concern countable point families, and their proofs, which use only lengths, fits, and the finiteness of the set of programs of each length.

The semimeasure inequalities in (b). $P_T(\emptyset) = \mathbf{1}[B \cup T$ consistent$] \le 1$. Let $\mathcal{Q} \notin \operatorname{dom} D$ and $P_T(D) > 0$. If T proves an answer a_0, every other answer is refuted by T and contributes 0, while a_0 adds no cost and keeps consistency; so the sum is $P_T(D)$. If T proves no answer, each of the at most $|\mathcal{Q}|$ answers consistent with $B \cup T \cup D^{\pm}$ contributes at most $2^{-\gamma(\mathcal{Q})}P_T(D) = P_T(D)/|\mathcal{Q}|$. If $P_T(D) = 0$, every term is 0.

Not lower semicomputable. Take $B = \emptyset$ and a language with a binary relation symbol. If $T \mapsto P_T(\emptyset)$ were lower semicomputable uniformly in T, the set of finite theories with $P_T(\emptyset) > \frac{1}{2}$, that is the consistent ones, would be r.e. The inconsistent ones are r.e. as well, so consistency of finite theories would be decidable, contrary to Church's theorem (standard; Church, 1936). □

Proposition A.8 (Coding bound for laws with literal exceptions)**.** *Let $n := |D|$.*

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $\|T\| + n + \mathrm{Exc}^Q_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$.

(b) Suppose that provability of answers from T is decidable uniformly in T, for T ranging over a class of presented theories with $K(T) \le \|T\| + c$ (finite theories over a decidable complete B, or pointwise interval laws). Then $\|T\| + \mathrm{Exc}^Q_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$ and $\|T\| + \mathrm{Exc}^R_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$ for every member T consistent with D.

Proof. (a) For a question $\mathcal{Q}$ put

$$w_T(a \mid \mathcal{Q}) := \tfrac{1}{2}\mathbf{1}[a \text{ is the first answer of } \mathcal{Q} \text{ found provable from } B \cup T] + \tfrac{1}{2}|\mathcal{Q}|^{-1},$$

where "first" refers to a fixed dovetailed proof search. Then $\sum_a w_T(a \mid \mathcal{Q}) \le 1$, and w_T is lower semicomputable uniformly in $(T, \mathcal{Q})$, because the first answer found never changes. So $P'_T(y) := \prod_j w_T(y_j \mid \mathcal{Q}_j)$ is a semimeasure on the answer lists for $\operatorname{dom} D$, lower semicomputable uniformly in $(T, \operatorname{dom} D)$. If $B \cup T$ is consistent and proves y_j, it proves no other answer of $\mathcal{Q}_j$, so y_j is the first found and $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}$; otherwise $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}|\mathcal{Q}_j|^{-1}$. Hence $-\log P'_T(y_D) \le n + \mathrm{Exc}^Q_D(T)$. The conditional coding theorem ($K(y \mid z) \le -\log P(y \mid z) + c_P$ for a semimeasure $P(\cdot \mid z)$ lower semicomputable uniformly in z; standard, see Li and Vitányi, 2008) gives $K(y_D \mid T, \operatorname{dom} D) \le n + \mathrm{Exc}^Q_D(T) + c$, and $K(y_D \mid \operatorname{dom} D) \le K(T) + K(y_D \mid T, \operatorname{dom} D) + c \le \|T\| + n + \mathrm{Exc}^Q_D(T) + c'$.

(b) With decidable provability put $w_T(a \mid \mathcal{Q}) := 1$ if a is the first answer, in a fixed order of $\mathcal{Q}$, that T proves; $:= 0$ for the other answers that T proves; and $:= |\mathcal{Q}|^{-1}$ for every answer if T proves none. This sums to at most 1 for every T, consistent or not, and it is computable. For T consistent with D the cost bound of (a) holds without the flag bit, and the same coding argument gives the first inequality. For code R, $P_T^R(\cdot \mid \operatorname{dom} D)$ is itself a semimeasure on answer lists, computable uniformly in $(T, \operatorname{dom} D)$ (Proposition A.7(a)), and the same argument gives the second. □

So the right line of slope 1 around a sufficient law (Hänni and Claude (Anthropic), 2026, Corollary 5.5(a)) holds in both classes; in code Q without decidability it carries the flag cost n, which makes it useless, and for that reason the analyses of Section 1 use laws whose provability on the data is decidable.

A.3 Deterministic theories and precision

Proposition A.9 (Compatibility is monotone in logical strength)**.**

(a) Let $B \cup T$ be consistent and $\mathrm{Cn}(T) \supseteq \mathrm{Cn}(T')$. On every data set with which T' is compatible ($B \cup T'$ consistent and proving every datum), T is compatible. So among the deterministic hypotheses of W_A and M_A the posterior odds of T against T' equal their prior odds on every data set with which T' is compatible, and T' is incompatible whenever T is: data never move the odds towards the weaker theory.

(b) $T_0 \vdash T^\varepsilon$ for every $\varepsilon \geq 0$, so every data set compatible with the honest law T^ε is compatible with the exact law T_0, at every sample size and every precision. With prior weights $2^{-\|T\|}$ in a code in which the tolerance clause costs at least one bit, the posterior odds of T_0 against T^ε are at least $2^{\|T^\varepsilon\| - \|T_0\|} \geq 2$ on every such data set.

(c) If $\sup_t |e(t)| > 0$, then T_0 is false and, on a question at precision $\delta' < \sup_t |e(t)|$, proves a false answer at some times; T^ε with $\varepsilon \geq \sup_t |e(t)|$ is true and is silent wherever its band straddles a cell boundary.

Proof. A consistent theory that proves everything T' proves proves every datum that T' proves; this is (a), and (b) is (a) with $T = T_0$, $T' = T^\varepsilon$. (c) is immediate from the definitions. □

Proposition A.10 (Precision monotonicity)**.** *If $B \cup T$ proves "$q(t) \in [v - \delta', v + \delta']$", then it proves "$q(t) \in [v - \delta, v + \delta]$" for every $\delta \geq \delta'$. So compatibility with data at a fixed set of centres is monotone in the precision.*

Proof. The first interval is contained in the second. □

Proposition A.11 (Guaranteed tolerances)**.** *In regime G:*

(a) T_0 proves datum j iff $|e(t_j) - \zeta_j| \leq \delta$, and refutes it otherwise.

(b) If $\delta \geq \varepsilon^* + \delta_0$, then T_0 is compatible, at every n, with every such data set whose errors lie in the essential range (so almost surely).

(c) Suppose the pairs $(e(t_j), \zeta_j)$ are i.i.d., $e(t_j)$ is independent of ζ_j, both $\pm\delta_0$ lie in the support of the law of ζ_j, and $\delta < \varepsilon^* + \delta_0$. Then the per-datum refutation probability $r(\delta) := \mathbb{P}[|e - \zeta| > \delta]$ is positive, T_0 survives n data with probability $(1 - r(\delta))^n$, and it is almost surely refuted eventually.

So the switch point $\delta = \varepsilon^* + \delta_0$ is sharp.

Proof. (a) T_0 proves $q(t_j) = f(t_j) = v_j + e(t_j) - \zeta_j$. This point lies in $[v_j - \delta, v_j + \delta]$ iff $|e(t_j) - \zeta_j| \le \delta$, and if it does not, T_0 proves the negation of the datum. (b) $|e - \zeta| \le |e| + |\zeta| \le \varepsilon^* + \delta_0$ almost surely. (c) Pick $\varepsilon' < \varepsilon^*$ and $s > 0$ with $\varepsilon' + \delta_0 - s > \delta$. Since ε^* is the essential supremum of $|e|$, $\mathbb{P}[e > \varepsilon'] > 0$ or $\mathbb{P}[e < -\varepsilon'] > 0$. In the first case $\mathbb{P}[\zeta < -\delta_0 + s] > 0$ (support), and independence gives $\mathbb{P}[e - \zeta > \delta] > 0$; the second case is symmetric, with $\zeta > \delta_0 - s$. Refutations of distinct data are independent, so survival has probability $(1 - r(\delta))^n \to 0$, and a refuted theory stays refuted. □

Proof of Proposition 1.2. T_0 proves "$q(t_j)$ lies in the cell of $f(t_j)$" and so refutes every other cell. Two points at distance $|e| < \Delta$ fall into different cells of a uniformly shifted grid of width Δ with probability $|e|/\Delta$, and at distance at least Δ they always do; this gives the conditional proof probability $(1 - |e(t_j)|/\Delta)^+$. The refutation indicators of distinct data are independent (they depend only on each datum's own error and dither), which gives the product $(1 - p_\Delta)^n$ and the geometric law of the first refutation; and $p_\Delta > 0$ iff $\mathbb{P}[e \neq 0] > 0$. For $\Delta \ge \varepsilon^*$, $\min(|e|/\Delta, 1) = |e|/\Delta$ almost surely. □

Proposition A.12 (Which exact theory a proof-requiring inducer settles on)**.** *Let $T_1, \dots, T_k$ be exact pointwise laws for the same quantity, with lengths $\|T_1\| < \cdots < \|T_k\|$ and errors $e_1, \dots, e_k$, and let the inducer choose the shortest compatible theory in this pool (the MAP of any prior that decreases with length, or K_A restricted to the pool). Let the data be i.i.d.*

(G) Under the hypotheses of Proposition A.11(c) at precision δ, almost surely the choice is eventually $T_{i^*(\delta)}$, $i^*(\delta) := \min\{i : \varepsilon_i^* + \delta_0 \le \delta\}$; if no i qualifies, every theory is eventually refuted. As δ decreases past the thresholds $\varepsilon_i^* + \delta_0$, the eventual choice moves to longer and more exact theories.

(R) In regime R, for every $\Delta > 0$, almost surely the choice is eventually the shortest T_i with $\mathbb{P}[e_i \neq 0] = 0$, whatever Δ is; if there is none, every theory is eventually refuted. At sample size n, T_i survives with probability $(1 - p_{\Delta,i})^n$, and for $\Delta \ge \varepsilon_i^*$ it is abandoned after about $\Delta/\mathbb{E}|e_i|$ data.

Proof. Apply Proposition A.11 and Proposition 1.2 to each T_i. The choice at n is the first survivor in order of length, and a finite pool has a last refutation time almost surely. □

Computations. In regime G, with $\varepsilon^* = 1'$, $\delta_0 = 0.5'$ and ζ uniform on $[-\delta_0, \delta_0]$, the refutation probability is exactly $r(\delta) = 0.218$ at $\delta = 1'$, 6.0×10^{-4} at $\delta = 1.49'$, and 0 for $\delta \ge 1.5'$, for a sinusoidal error and for an error equal to $\pm\varepsilon^*$ on 5% of the times and 0 otherwise. In regime R, simulated survival after 1, 10 and 100 data matches $(1 - p_\Delta)^n$ at every Δ from $0.5'$ to $600'$; for the sinusoid with $\varepsilon^* = 1'$ the mean first-refutation index is 94.1 against the exact $1/p_\Delta = \pi\Delta/2 = 94.25$ at $\Delta = 60'$, and 942.7 against 942.48 at $\Delta = 600'$. For the pool T_1 (length 100, $\varepsilon^* = 10'$), T_2 (length 150, $\varepsilon^* = 1'$), T_3 (length 400, exact), with sinusoidal errors and 2000 runs, every run chooses the predicted theory at $n = 10^4$ in regime G with $\delta_0 = 0.5'$ at each of $\delta = 20'$, $10.5'$, $5'$, $1.5'$, $1.2'$ (predictions T_1, T_1, T_2, T_2, T_3); in regime R every run ends at T_3, T_1 being abandoned after about 94 data at $\Delta = 10^\circ$ and 9.4 at $\Delta = 1^\circ$, and T_2 after about 942 and 94.

A.4 Honest laws, laws with exceptions, laws with error models

Forms and pricings. Work in regime R. The forms are: the empty law $\emptyset$; the *patched tolerance law* $T^{\varepsilon,-Z}$, the tolerance law T^ε withdrawn at the set Z of times of the data it refutes (fewer patches

are inconsistent with the data, more only add cost); for $\varepsilon = 0$ the exact law, written T_0 while it has no patch and T_0^{-Z} after; for $\varepsilon \geq \varepsilon^*$ the honest law T^ε, which is true and never patched; and a true exact theory T_{tr}. Their lengths, relative to a common part, are $\|\emptyset\| = 0$, $\|T_0\| = \ell_0$, $\|T^\varepsilon\| = \ell_0 + h$ (h the length of the tolerance clause and its numeral) and $\|T_{\mathrm{tr}}\| = \ell_0 + h_{\mathrm{tr}}$. A pricing fixes the price γ_{si} of a datum on which a form is silent and the price γ_{ex} of an exception:

- *Q/det*: code Q, each patch naming its time inside the law at c_{time} bits; $\gamma_{\mathrm{si}} = \gamma$ and $\gamma_{\mathrm{ex}} = \lambda c_{\mathrm{time}} + \gamma$. This is a finite-horizon cost model (Proposition A.15).
- *Q/rand*: code Q with randomised exceptions: a theory machine withdraws the law at each time independently with probability ρ, and the fit is charged $-\log(1-\rho)$ per kept datum and $-\log\rho + \gamma$ per exception. The share ρ can be stated in $O(\log n)$ bits or integrated out: drawing ρ uniformly from the tape and then withdrawing independently is a single theory machine, whose cost for k exceptions among n data is $\log\left((n+1)\binom{n}{k}\right) + k\gamma$. With $\rho = p_\Delta$, or integrated out, the rate is $H_2(p_\Delta) + p_\Delta\gamma$, where H_2 is the binary entropy, so $\gamma_{\mathrm{ex}} = \gamma + H_2(p_\Delta)/p_\Delta$. These are exact theory-machine costs for every n.
- *R/det* and *R/rand*: as above, with a silent datum priced by random completion inside the band (code R); an exception of a patched law is priced γ in code R too (Proposition A.3(c)).

SAI's pricing is R/rand: the honest law with random completion and the exact law with random exceptions are theory machines, and R/rand compares their weights in M_A's mixture.

Lemma A.13 (What a tolerance law does with one datum)**.** *In regime R, fix a datum with error e and let $\varepsilon \geq 0$. Over the dither, T^ε refutes the datum with probability* $\min\left((|e| - \varepsilon)^+/\Delta, 1\right)$*, proves it with probability* $\left(1 - (\varepsilon + \max(|e|, \varepsilon))/\Delta\right)^+$*, and is silent otherwise. In the* coarse regime $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ *it is silent with probability exactly $2\varepsilon/\Delta$, whatever e is, and refutes with probability $(|e| - \varepsilon)^+/\Delta$. For $\varepsilon = 0$ it refutes with probability* $\min(|e|/\Delta, 1)$ *and is never silent.*

Proof. Place q at 0. The cell containing it is $[-u\Delta, (1-u)\Delta)$ with u uniform on $[0,1)$, and the band is $[e - \varepsilon, e + \varepsilon]$, since $f = q + e$; by symmetry let $e \geq 0$. *Refutation* means that the band misses the cell. If $e \leq \varepsilon$, the band contains 0, so this is impossible. If $e > \varepsilon$, the band lies in $(0, \infty)$, and it misses the cell iff $(1-u)\Delta \leq e - \varepsilon$, of probability $\min((e-\varepsilon)/\Delta, 1)$. *Proof of the datum* means that the band lies in the cell: $-u\Delta \leq e - \varepsilon$ and $e + \varepsilon < (1-u)\Delta$. If $e \leq \varepsilon$, this is $u \in [(\varepsilon - e)/\Delta, (\Delta - \varepsilon - e)/\Delta)$, of probability $(1 - 2\varepsilon/\Delta)^+$; if $e > \varepsilon$, the first condition holds automatically and the second is $u < (\Delta - e - \varepsilon)/\Delta$. Both cases equal $(1 - (\varepsilon + \max(e, \varepsilon))/\Delta)^+$. When $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ the positive parts are inactive for every e in the essential range, and silence has probability $1 - (1 - (\varepsilon + \max(e, \varepsilon))/\Delta) - (\max(e, \varepsilon) - \varepsilon)/\Delta = 2\varepsilon/\Delta$. □

The coarse-regime condition cannot be weakened to $\Delta \geq \varepsilon + \varepsilon^*$ when $\varepsilon > \varepsilon^*$: with $|e| = \varepsilon^* = 1$, $\varepsilon = 2$ and $\Delta = 3$ the formula $2\varepsilon/\Delta$ would give a silence probability of $4/3$.

Proposition A.14 (Rates)**.** *In regime R, write $z_j := (e_j, u_j)$. Each form X has $J_\lambda(X; D_n) = a_X + \sum_{j \leq n} c_X(z_j)$ with a constant a_X and i.i.d. per-datum costs $c_X(z_j)$ (under Q/rand and R/rand with ρ stated; integrating ρ out adds $O(\log n)$). In the coarse regime $\Delta \geq 2\varepsilon^*$ the rates $m_X := \mathbb{E}c_X$ are: $m_\emptyset = \gamma$; $m_{T_{\mathrm{tr}}} = 0$; $m_{T_0} = \infty$ if $\mathbb{P}[e \neq 0] > 0$ (an unpatched exact law is eventually refuted); $m_{T_0^{-}z} = p_\Delta\gamma_{\mathrm{ex}}$, that is $p_\Delta(\lambda c_{\mathrm{time}} + \gamma)$ under the det pricings and $H_2(p_\Delta) + p_\Delta\gamma$ under the rand pricings, with $p_\Delta = \mathbb{E}|e|/\Delta$; and $m_{T^{\varepsilon^*}} = s_\Delta\gamma_{\mathrm{si}}$ with $s_\Delta = 2\varepsilon^*/\Delta$, that is $s_\Delta\gamma$ in code Q and $s_\Delta\beta_e$ in code R, β_e as in Lemma A.18.*

Proof. For the patched exact law at its cheapest patch set, a datum costs γ_{ex} if T_0 refutes it and 0 otherwise: under det the patch names the time ($\lambda c_{\mathrm{time}}$) and the datum is then literal (γ); under

rand it is the flag code. Lemma A.13 with $\varepsilon = 0$ gives the probability p_Δ. The honest law costs γ_{si} on a datum iff it is silent, which in the coarse regime has probability $2\varepsilon^*/\Delta$ independently of e (Lemma A.13); in code R the expected cost given silence, averaged over the errors, is β_e (Lemma A.18(i)). □

Proposition A.15 (The price of naming exceptions deterministically)**.** *In regime R, let $Z_n \subseteq \{1, \dots, n\}$ be the indices of the data that the exact law refutes, so that the events $j \in Z_n$ are independent with probability $p_\Delta \in (0, 1)$.*

(a) For every prefix-free code of the finite subsets of $\{1, \dots, n\}$, even one that depends on n, the expected code length of Z_n is at least $nH_2(p_\Delta)$; so the expected cost per exception is at least $H_2(p_\Delta)/p_\Delta$ bits, the cost of randomised exceptions.

(b) If the patched law names each patched time t_j by a numeral in a prefix-free code ℓ_{num} of numerals, then $\mathbb{E}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq p_\Delta n \log n$, so the cost per exception is at least $\log n$ on average. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$ for all j (any standard numeral code), the cost per datum $\frac{1}{n}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j)$ tends to ∞ almost surely.

(c) Hence Q/det and R/det with a constant c_{time} are cost models valid only while the times used cost about c_{time} bits to name, that is for $n \lesssim 2^{c_{\mathrm{time}}}$. Asymptotically, at fresh times, deterministic patching costs at least as much per exception as randomised exceptions, and literal patching costs unboundedly more; so the patched exact law under det is eventually beaten by the exact law with random exceptions.

Proof. (a) Let P_Z be the law of Z_n and ℓ' a prefix-free code. By Kraft's inequality $\sum_Z 2^{-\ell'(Z)} \leq 1$, and by Jensen's inequality, with H the entropy,

$$\mathbb{E}\ell'(Z_n) - H(Z_n) = -\sum_Z P_Z(Z) \log \frac{2^{-\ell'(Z)}}{P_Z(Z)} \geq -\log \sum_Z 2^{-\ell'(Z)} \geq 0,$$

where $H(Z_n) = nH_2(p_\Delta)$. The expected number of exceptions is $p_\Delta n$. (b) The numerals $t_1, \dots, t_n$ are distinct, so Kraft gives $\sum_{j\leq n} 2^{-\ell_{\mathrm{num}}(t_j)} \leq 1$, and Jensen gives $2^{-\frac{1}{n}\sum_j \ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}\sum_j 2^{-\ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}$, that is $\sum_{j\leq n} \ell_{\mathrm{num}}(t_j) \geq n \log n$. Since $\mathbb{P}[j \in Z_n] = p_\Delta$ for every j, $\mathbb{E}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq p_\Delta n \log n$. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$, then $\frac{1}{n}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq \frac{1}{n}\#\{j \in Z_n : j > n/2\}(\log n - 1 - c)$, and the count over n tends to $p_\Delta/2$ almost surely by the strong law of large numbers. (c) follows from (a), (b) and Proposition A.14. □

A computation (code in the accompanying repository) gives, at $p_\Delta = 0.01$, 15.0, 23.0 and 30.3 bits per exception for literal listing by Elias delta numerals at $n = 10^3$, 10^5 and 10^7, and 9.02, 8.13 and 8.08 bits for an enumerative code of the set, against $H_2(p_\Delta)/p_\Delta = 8.08$.

Proposition A.16 (The peakedness criterion)**.** *Let $\Delta \geq 2\varepsilon^*$, under any of the four pricings (the deterministic ones as a cost model).*

(a) The honest law T^{ε^*} has a lower rate than the exact law with exceptions T_0^{-Z} iff $\mathrm{pk}(e) < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

(b) If the exact law with exceptions comes from a law f_1 with error e_1, and the honest law from a law f_2 of the same length up to $O(1)$, with tolerance $\varepsilon_2 \geq \operatorname{ess\,sup}|e_2|$ and $\Delta \geq 2\varepsilon_2$, then the honest law has the lower rate iff $\varepsilon_2/\mathbb{E}|e_1| < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

Proof. By Proposition A.14, $s_\Delta\gamma_{\mathrm{si}} < p_\Delta\gamma_{\mathrm{ex}}$ iff $2\varepsilon^*\gamma_{\mathrm{si}} < \mathbb{E}|e|\,\gamma_{\mathrm{ex}}$. In (b), $s_\Delta = 2\varepsilon_2/\Delta$ (Lemma A.13) and $p_\Delta = \mathbb{E}|e_1|/\Delta$. □

Under Q/det the criterion reads $\lambda c_{\text{time}} > \gamma(2\text{pk}(e) - 1)$. Simulated rates for a sinusoid ($\text{pk} = \pi/2$) with $c_{\text{time}} = 22$ pick the exact law with exceptions at $\lambda = 1$ and the honest law at $\lambda = 2$ for $\Delta = 2'$, $3'$ and $10'$, and the honest law at both for $\Delta = 60'$, as the threshold $(\lambda c_{\text{time}} + \gamma)/(2\gamma)$ predicts; for an error with $\text{pk} = 20$ they pick the exact law with exceptions everywhere. Under Q/rand it reads $\text{pk}(e) < \frac{1}{2} + H_2(p_\Delta)/(2p_\Delta\gamma)$; since $H_2(p)/p \approx \log(\text{e}/p)$ for small p, the threshold grows only logarithmically in $1/p_\Delta$: at $\gamma = \log 360$ it is 1.17 at $p_\Delta = 10^{-3}$, 1.76 at 10^{-6} and 2.93 at 10^{-12}. For the sinusoid it is 0.637, 0.742 and 0.971 at $\Delta = 3'$, $10'$ and $60'$, so the exact law with random exceptions wins although the error is spread. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand the threshold is $(\gamma + H_2(p_\Delta)/p_\Delta)/(2\beta_e)$, roughly 5 to 30 for typical cell widths (9.2–9.3 for the sinusoid, 20.4–20.5 for the error with $\text{pk} = 20$). This comparison is relative to its pool: the exact law with random exceptions completes its exceptions over the whole circle, while the equally short two-point completion $\text{HY}(\rho)$ completes them inside the band, and against it the error with $\text{pk} = 20$ is held as "exact except in about 10% of cases" at a third of the uniform band's rate (Proposition 1.3(d) and the computation after its proof).

Proposition A.17 (Optimal tolerance)**.** *Under Q/det in the regime $\Delta \geq 2\varepsilon^*$, the patched tolerance law $T^{\varepsilon,-Z}$ with $0 \leq \varepsilon \leq \varepsilon^*$ has rate*

$$\bar{m}(\varepsilon) = \frac{2\varepsilon\gamma + (\lambda c_{\text{time}} + \gamma)\,\mathbb{E}(|e| - \varepsilon)^+}{\Delta}.$$

$\bar{m}$ is convex, with right derivative $\big(2\gamma - (\lambda c_{\text{time}} + \gamma)\mathbb{P}[|e| > \varepsilon]\big)/\Delta$, and its minimisers are the ε with $\mathbb{P}[|e| > \varepsilon] \leq \theta_{\text{ex}} \leq \mathbb{P}[|e| \geq \varepsilon]$, where $\theta_{\text{ex}} := 2\gamma/(\lambda c_{\text{time}} + \gamma)$. So the optimal law states the $(1 - \theta_{\text{ex}})$-quantile of $|e|$ as its tolerance and lists the remaining share θ_{ex} of cases as exceptions. If $\theta_{\text{ex}} \geq 1$ ($\lambda c_{\text{time}} \leq \gamma$), the exact law with patches is optimal. The pure honest law ($\varepsilon = \varepsilon^*$, no exceptions) is optimal iff $\mathbb{P}[|e| \geq \varepsilon^*] \geq \theta_{\text{ex}}$, which never holds for an error law without an atom at its maximum.

Proof. A datum costs γ if it is not proved, and in addition λc_{time} if it is refuted (the patch). By Lemma A.13, with $\Delta \geq 2\varepsilon^* \geq \varepsilon + \varepsilon^*$, $\mathbb{P}[\text{not proved}] = (2\varepsilon + \mathbb{E}(|e| - \varepsilon)^+)/\Delta$ and $\mathbb{P}[\text{refuted}] = \mathbb{E}(|e| - \varepsilon)^+/\Delta$, which gives $\bar{m}$. The function $\varepsilon \mapsto \mathbb{E}(|e| - \varepsilon)^+$ is convex, with right derivative $-\mathbb{P}[|e| > \varepsilon]$ and left derivative $-\mathbb{P}[|e| \geq \varepsilon]$. This gives convexity and the derivatives of $\bar{m}$, and a convex function is minimised exactly where its right derivative is non-negative and its left derivative non-positive, which is the stated condition. The last two claims are the cases $\varepsilon = 0$ and $\varepsilon = \varepsilon^*$ of it. □

Minimising $\theta\varepsilon + \mathbb{E}(|e| - \varepsilon)^+$ over ε is the quantile-loss fact, and in ν-support-vector regression the parameter ν likewise bounds the share of points outside an ε-tube whose width is a quantile of the residuals (cf. Schölkopf et al., 2000). What is specific here is the reading for axiom inducers: the tube is a tolerance clause, the points outside it are patched exceptions, and θ_{ex} is a ratio of literal costs. The steeper the prior, or the dearer the naming of an exception, the smaller θ_{ex}: the law states a wider tolerance and lists fewer exceptions. A computation at $\Delta = 10'$, $c_{\text{time}} = 22$ finds the grid minimiser at 0.52, 0.80 and 0.94 for a sinusoid of amplitude 1 at $\lambda = 1, 2, 4$, against the predicted quantiles 0.496, 0.807 and 0.939.

Lemma A.18 (The code-R cost of the honest law, exactly)**.** *Let $\varepsilon \geq \varepsilon^*$ and let $w := (\varepsilon - e)/(2\varepsilon) \in [0, 1]$ be the position of the true value in the band. Put $\vartheta(z) := 1 - z + z\ln z$ for $0 \leq z < 1$ ($\vartheta(0) = 1$)*

and $\vartheta(z) := 0$ *for* $z \geq 1$. *The expected code-R cost of a datum for* T^ε, *over the dither, is*

$$\bar{c}_\Delta(w) = \begin{cases} \dfrac{2\varepsilon}{\Delta}\big(\log \mathrm{e} - H_2(w)\big) & \text{if } \Delta \geq 2\varepsilon, \\ \log \dfrac{2\varepsilon}{\Delta} + \log \mathrm{e}\left[\vartheta\Big(\dfrac{2\varepsilon w}{\Delta}\Big) + \vartheta\Big(\dfrac{2\varepsilon(1-w)}{\Delta}\Big)\right] & \text{if } \Delta \leq 2\varepsilon. \end{cases}$$

(i) In the coarse regime the cost given silence is $\log \mathrm{e} - H_2(w)$. *Averaged over the errors,* $\gamma_{\mathrm{si}} = \beta_e := \log \mathrm{e} - \mathbb{E}H_2(w) \in [\log \mathrm{e} - 1, \log \mathrm{e}] = [0.443, 1.443]$ *bits.*

(ii) In the fine regime T^ε *is silent on every datum, and* $\log(2\varepsilon/\Delta) \leq \bar{c}_\Delta(w) \leq \log(2\varepsilon/\Delta) + \log \mathrm{e}$; *the upper bound is approached as* $w \to 0$ *at* $\Delta = 2\varepsilon$.

(iii) For every w, $\bar{c}_\Delta(w)$ *is continuous and strictly decreasing in* $\Delta \in (0, \infty)$.

Proof. Coarse regime. The band straddles a cell boundary with probability $2\varepsilon/\Delta$ (Lemma A.13, since $\varepsilon \geq \varepsilon^*$); otherwise it lies in a cell and the datum costs 0. Given a straddle, exactly one boundary lies in the band, at a relative position v uniform on $[0, 1]$ (a uniformly shifted lattice of spacing at least the band's length). The true cell contains the part of the band on the true value's side, of relative length v if $w < v$ and $1 - v$ otherwise, and random completion inside the band puts that much probability on it. So the expected cost given a straddle is, in nats,

$$\int_w^1 -\ln v \, dv + \int_0^w -\ln(1-v)\, dv = 1 + w \ln w + (1-w)\ln(1-w),$$

which is $\log \mathrm{e} - H_2(w)$ in bits.

Fine regime. Let y be the length of the part of the true cell inside the band; the cost is $\log(2\varepsilon/y) = \log(2\varepsilon/\Delta) - \log(y/\Delta)$. With $a := 2\varepsilon w/\Delta$ and $b := 2\varepsilon(1-w)/\Delta$, so that $a + b = 2\varepsilon/\Delta \geq 1$, the band is $[-a\Delta, b\Delta]$ around the true value, and with dither u the true cell is $[-u\Delta, (1-u)\Delta)$, so $y/\Delta = \min(u, a) + \min(1-u, b)$. If $a, b \geq 1$, then $y = \Delta$ and the excess $-\ln(y/\Delta)$ is 0. If $a < 1 \leq b$, then $y/\Delta = 1$ for $u < a$ and $1 + a - u$ for $u \geq a$, so the expected excess is $\int_a^1 -\ln(1 + a - u)\, du = \int_a^1 -\ln s \, ds = \vartheta(a)$; the case $b < 1 \leq a$ is symmetric. If $a, b < 1$, then $1 - b \leq a$, and $y/\Delta = u + b$ for $u < 1 - b$, 1 on $[1 - b, a]$, and $a + 1 - u$ for $u > a$; the expected excess is $\int_b^1 -\ln s \, ds + \int_a^1 -\ln s \, ds = \vartheta(a) + \vartheta(b)$.

(i) is the coarse case, and $\mathbb{E}H_2(w) \in [0, 1]$. (ii) In the fine regime $2\varepsilon \geq \Delta$, so T^ε is silent on every datum by Lemma A.13. $\vartheta \geq 0$ and ϑ is decreasing on $[0, 1]$ ($\vartheta'(z) = \ln z$). In the last case $\vartheta(a) + \vartheta(b)$ is largest on $a + b = 1$, where it equals $1 - H_2(a)\ln 2 \leq 1$; in the middle case $\vartheta(a) \leq \vartheta(0) = 1$. The value 1 is approached as $a \to 0$, $b \to 1$. (iii) Write $r := 2\varepsilon/\Delta$ and the cost in nats. In the coarse regime ($r \leq 1$) it is $r(1 - H_2(w)\ln 2)$, increasing in r because $1 - H_2(w)\ln 2 \geq 1 - \ln 2 > 0$. In the fine regime ($r \geq 1$) it is $\ln r + \vartheta(wr) + \vartheta((1-w)r)$, and on each piece its derivative in r is $\frac{1}{r}(1 + \sum z \ln z)$, the sum over $z \in \{wr, (1-w)r\}$ with $z < 1$; since $z \ln z \geq -1/\mathrm{e}$, this is at least $\frac{1}{r}(1 - 2/\mathrm{e}) > 0$. The cost is continuous in r (ϑ is continuous at 1), and at $r = 1$ the two forms agree, since $\vartheta(w) + \vartheta(1-w) = 1 - H_2(w)\ln 2$. So it is strictly increasing in r, that is strictly decreasing in Δ. □

A computation (code in the accompanying repository) confirms the closed form against a 5000-point quadrature over the dither to 10^{-4} bits on 36090 cells, with largest fine-regime excess $1.442695 = \log \mathrm{e}$; the coarse averages are $\beta_e = 0.885$ for a sinusoid and 0.493 for an error equal to $\pm\varepsilon^*$ on 5% of the times.

Band-confined completions. The machine $\mathrm{BC}_{\mathcal{G}}$ of Section 1 draws $\hat{w}_i \sim \mathcal{G}$ by the inverse distribution function from the uniform variables U_i of the proof of Proposition A.3, and emits the value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$ by narrowing intervals for the continuous part of $\mathcal{G}$ and exactly at an atom. A computable $\mathcal{G}$ is, for instance, piecewise linear with rational knots plus finitely many atoms at rational points. The argument of Proposition A.3(a) gives the claims made in Section 1: $K(\mathrm{BC}_{\mathcal{G}}) \le \|T^{\varepsilon}\| + K(\mathcal{G}) + c$, the random theory is always consistent, and on data at fresh times $\nu_{\mathrm{BC}_{\mathcal{G}}}(D) = \prod_j \mathcal{G}(C'_j)$ (cell endpoints are null events for continuous dithers).

Proof of Proposition 1.3. (a) As in Lemma A.18: a cell boundary falls into the band with probability $2\varepsilon/\Delta$, at a relative position v uniform on $[0,1]$ and independent of w. With no boundary in the band the cell contains the band and the datum costs 0. With a boundary at v, the true cell contains $[0,v)$ if $w < v$, of $\mathcal{G}$-mass $\mathcal{G}(v)$, and $[v,1]$ if $w > v$, of mass $1 - \mathcal{G}(v)$; this gives $c_{\mathcal{G}}$, which does not depend on Δ.

(b) Given v, the event $w < v$ has probability $\mathcal{F}(v)$, so the conditional expected cost is $\mathcal{F}(v)(-\log \mathcal{G}(v)) + (1-\mathcal{F}(v))(-\log(1-\mathcal{G}(v))) \ge H_2(\mathcal{F}(v))$, with equality iff $\mathcal{G}(v) = \mathcal{F}(v)$ (Gibbs' inequality). Integrate over v. The uniform band has $\mathcal{G}(v) = v$, which equals $\mathcal{F}$ on $[0,1]$ iff w is uniform, that is iff e is uniform on $[-\varepsilon, \varepsilon]$.

(c) Let $P^{\star}$ be the true conditional law of the answer list given the question list. The errors are independent of the questions, and $q = f - e$, so the conditional probability of the observed cell of $q(t_j)$ is $\mathcal{F}(C'_j)$; hence $P^{\star}(D_n) = \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n) = \prod_{j \le n} \mathcal{F}(C'_j)$, and $-\log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$ is a sum of i.i.d. terms with finite mean $r_{\mathcal{F}}(\Delta)$; the strong law of large numbers gives the limit. Let G' be any theory machine. Answers to one question are pairwise B-incompatible, so for a fixed question list the events "the random theory is consistent and proves the answer list y" are disjoint across y; hence $\sum_y \nu_{G'}(y) \le 1$ and $\mathbb{E}_{P^{\star}}[\nu_{G'}(D_n)/P^{\star}(D_n) \mid \text{questions}] \le 1$. By Markov's inequality $\mathbb{P}[-\log \nu_{G'}(D_n) < -\log P^{\star}(D_n) - 2\log n] \le n^{-2}$, and by the Borel–Cantelli lemma, almost surely $-\log \nu_{G'}(D_n) \ge -\log P^{\star}(D_n) - 2\log n$ for all large n. Dividing by n gives the lim inf bound, and $G' = G_U$ gives it for L_A. If $\mathcal{F}$ is computable, dominance gives $L_A(D_n) \le K(\mathrm{BC}_{\mathcal{F}}) + c - \log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$, hence the limit. In the coarse regime $r_{\mathcal{F}}(\Delta) = (2\varepsilon/\Delta)c_{\mathcal{F}}$ by (a) and the equality case of (b).

(d) Write $\sigma := 1 - \rho$ for the weight of the atom at $\frac{1}{2}$. The family is symmetric, $\mathcal{G}(1-v) = 1 - \mathcal{G}(v)$, so the cost given w equals the cost given $1-w$, and we may take $w \le \frac{1}{2}$. *Convexity.* For fixed (w,v) the probability assigned to the true side is affine in σ and positive for $\sigma \in [0,1)$, and $-\ln$ of a positive affine function is convex; so $c_{\mathrm{HY}(\rho)}$, an expectation of convex functions, is convex in σ, hence in $\rho \in (0,1]$. *Right derivative at $\sigma = 0$, in nats.* For $v < w$ the true side has probability $1 - (1-\sigma)v$, and the derivative of $-\ln$ of it at $\sigma = 0$ is $-v/(1-v) \in [-1,0]$; for $w < v < \frac{1}{2}$ the probability is $(1-\sigma)v$, with derivative $+1$; for $v > \frac{1}{2}$ it is $\sigma + (1-\sigma)v$, with derivative $-(1-v)/v \in [-1,0]$. Difference quotients of convex functions decrease monotonically as $\sigma \downarrow 0$, so by monotone convergence the derivative of the expectation is the expectation of the derivatives:

$$c'(0^+) = \mathbb{E}\Big[-\int_0^w \tfrac{v}{1-v}\,dv + (\tfrac{1}{2} - w) - \int_{1/2}^1 \tfrac{1-v}{v}\,dv \Big] = \mathbb{E}\big[\ln(1-w) + 1 - \ln 2\big],$$

using $\int_0^w \frac{v}{1-v}dv = -\ln(1-w) - w$ and $\int_{1/2}^1 \frac{1-v}{v}dv = \ln 2 - \frac{1}{2}$. For $w \le \frac{1}{2}$, $1 - w = (1 + |e|/\varepsilon)/2$, so $c'(0^+) = \mathbb{E}\ln(1 + |e|/\varepsilon) + 1 - 2\ln 2$. *The criterion.* If $c'(0^+) < 0$, some $\sigma > 0$ beats $\sigma = 0$, which is the uniform band; if $c'(0^+) \ge 0$, convexity gives $c(\sigma) \ge c(0)$ for every σ. *Necessity.* $\ln(1+z) \ge z\ln 2$ on $[0,1]$ (a concave function lies above its chord), so $\mathbb{E}\ln(1+|e|/\varepsilon) \ge \ln 2/\mathrm{pk}_{\varepsilon}(e)$, and $c'(0^+) < 0$ forces $\mathrm{pk}_{\varepsilon}(e) > \ln 2/(2\ln 2 - 1)$. *Sufficiency.* Jensen's inequality gives

$\mathbb{E}\ln(1+|e|/\varepsilon) \le \ln(1+1/\mathrm{pk}_\varepsilon(e))$, which is below $2\ln 2-1$ iff $\mathrm{pk}_\varepsilon(e) > \mathrm{e}/(4-\mathrm{e})$. *Boundary.* $\int_0^1 \ln(1+z)\,dz = 2\ln 2 - 1$. □

The following computation (synthetic errors, 2×10^5 samples, band $\varepsilon = \varepsilon^*$; code in the accompanying repository) gives per-straddle costs in bits in the coarse regime. SH is a 40-bin smoothed histogram fitted on a random half of the samples and evaluated on the other half, $c_{\mathcal{F}}$ is the bound of Proposition 1.3(b) on that half, and $c'(0^+)$ (in nats) is the derivative that decides whether some HY(ρ) with $\rho < 1$ beats the uniform band.

error	pk	uniform	best HY	ρ	SH	$c_{\mathcal{F}}$	$c'(0^+)$
sinusoid	1.57	0.8854	0.8854	1	0.8514	0.8510	$+0.087$
$\pm\varepsilon^*$ on 5%	20.1	0.4924	0.1790	0.099	0.2006	0.1704	-0.352
$\lvert e\rvert$ uniform	2.00	0.7213	0.7213	1	0.7227	0.7227	≈ 0
0.25 or 1 (90/10%)	3.08	0.5832	0.5543	0.68	0.4797	0.4640	-0.116

So the two-point form beats the uniform band exactly when $c'(0^+) < 0$, and the fitted histogram comes within about 0.02 bits of the Gibbs bound of the half it did not see.

Proposition A.19 (Fine data)**.** *Let $\Delta \le 2\varepsilon^*$. The honest law T^{ε^*} is silent on every datum.*

(a) In code Q its rate is $\gamma = m_\emptyset$, so it never beats the empty law, and T_0^{-Z} beats the empty law iff $p_\Delta \gamma_{\mathrm{ex}} < \gamma$.

(b) In code R its rate is at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$, which is below $m_\emptyset = \gamma = \log(\mathsf{R}/\Delta)$ by at least $\log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e}$ bits per datum; this is positive iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$, about 66.2° for a longitude.

(c) If moreover $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$ and $\mathbb{P}[|e| < \Delta]\log(1/\Delta) \to 0$ as $\Delta \to 0$ (for instance, if $|e|$ has a bounded density near 0), then in code R the honest law has the least rate among $\emptyset$, T_0^{-Z} and T^{ε^*} for all small Δ.

Proof. Silence and (a) are Lemma A.13 and Proposition A.14's argument with silence probability 1. (b) is Lemma A.18(ii). (c) $m_{T_0^- z} \ge p_\Delta \gamma$, and $1 - p_\Delta = \mathbb{E}(1-|e|/\Delta)^+ \le \mathbb{P}[|e| < \Delta]$, so $m_{T_0^- z} - m_{T^{\varepsilon^*}} \ge \log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e} - \mathbb{P}[|e| < \Delta]\log(\mathsf{R}/\Delta)$, which is positive for small Δ; and (b) handles the empty law. □

So under SAI's pricing an approximate law keeps its value at precisions finer than its error: it narrows each answer to its band, which is worth $\log(\mathsf{R}/2\varepsilon^*)$ bits per datum (about 13 bits for a $1'$ error on a full circle of longitude), and by Proposition 1.3 a completion shaped like the error narrows it further.

Theorem A.20 (Almost-sure selection)**.** *Let $\mathcal{X}$ be a finite set of forms with $J_\lambda(X; D_n) = a_X + \sum_{j\le n} c_X(z_j) + r_X(n)$, where the z_j are i.i.d., $c_X(z) \in [0,\infty]$, and $r_X(n) = o(n)$ almost surely ($r_X = 0$ for deterministic forms and for stated ρ; $O(\log n)$ when ρ is integrated out). Let $m_X := \mathbb{E}c_X \in [0,\infty]$, $m_{\min} := \min_X m_X < \infty$ and $\mathcal{X}^* := \{X : m_X = m_{\min}\}$.*

(i) Almost surely, for all large n every λ-MAP within $\mathcal{X}$ lies in $\mathcal{X}^*$. If $\mathcal{X}^* = \{X^*\}$, the λ-MAP is eventually X^* alone.

(ii) Let $X, Y \in \mathcal{X}^*$ with $r_X = r_Y = 0$. If $c_X - c_Y$ is almost surely finite, with finite variance, and not almost surely constant, then almost surely $J_\lambda(X; D_n) < J_\lambda(Y; D_n)$ for infinitely many n and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ for infinitely many n: there is no eventual winner. If $c_X - c_Y = 0$ almost surely, a_X against a_Y decides at every n.

(iii) Let $r_X = r_Y = 0$, $m_Y < m_X < \infty$, $\mathbb{E}|c_X - c_Y| < \infty$ and $x := a_Y - a_X > 0$. Let τ_x be the first n with $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$, ς_x the last n with $J_\lambda(Y; D_n) \geq J_\lambda(X; D_n)$, and $\mu := m_X - m_Y$. Then $\mathbb{E}\tau_x > x/\mu$; if moreover $c_X - c_Y \leq b$ almost surely, then $\mathbb{E}\tau_x \leq (x+b)/\mu$; and on almost every sample path, $\tau_x/x \to 1/\mu$ and $\varsigma_x/x \to 1/\mu$ as $x \to \infty$.

Proof. The probabilistic facts used are standard (see Durrett, 2019). (i) Let $X^* \in \mathcal{X}^*$; then $\frac{1}{n} J_\lambda(X^*; D_n) \to m_{\min}$ almost surely. Let $X \notin \mathcal{X}^*$. If $\mathbb{P}[c_X = \infty] > 0$, almost surely some datum has infinite cost, after which $J_\lambda(X; D_n) = \infty$. If $c_X < \infty$ almost surely and $m_X = \infty$, then for each K, $\frac{1}{n}\sum_{j \leq n} \min(c_X(z_j), K) \to \mathbb{E}\min(c_X, K)$, and $\mathbb{E}\min(c_X, K) \to \infty$ as $K \to \infty$, so $\frac{1}{n} J_\lambda(X; D_n) \to \infty$. If $m_X < \infty$, then $\frac{1}{n}(J_\lambda(X; D_n) - J_\lambda(X^*; D_n)) \to m_X - m_{\min} > 0$. Since $\mathcal{X}$ is finite, the claim follows. (ii) $S_n := \sum_{j \leq n}(c_X - c_Y)(z_j)$ is a random walk with mean-zero, non-degenerate increments of finite variance. For each K, $\mathbb{P}[S_n > K] \to \frac{1}{2}$ by the central limit theorem, so $\mathbb{P}[S_n > K \text{ infinitely often}] \geq \limsup_n \mathbb{P}[S_n > K] = \frac{1}{2}$. This event is invariant under finite permutations of the increments, so by the Hewitt–Savage 0–1 law its probability is 1. Intersecting over $K \in \mathbb{N}$ gives $\limsup_n S_n = +\infty$ almost surely, and likewise $\liminf_n S_n = -\infty$; so $J_\lambda(X; D_n) - J_\lambda(Y; D_n) = a_X - a_Y + S_n$ changes sign infinitely often. The degenerate case is immediate. (iii) $S_n := \sum_{j \leq n}(c_X - c_Y)(z_j)$ has i.i.d. integrable increments with mean $\mu > 0$, and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ iff $S_n > x$. So $\tau_x = \min\{n : S_n > x\}$ is a stopping time with $\mathbb{E}\tau_x < \infty$ (a random walk with positive drift), and Wald's identity gives $\mathbb{E}S_{\tau_x} = \mu\mathbb{E}\tau_x$ with $S_{\tau_x} > x$; if the increments are at most b, also $S_{\tau_x} \leq x + b$. For the path statement fix $\epsilon \in (0, \mu)$; by the strong law there is $N(\omega)$ with $|S_n - n\mu| \leq \epsilon n$ for $n \geq N$. For $x > \max_{n \leq N} S_n$ we have $\tau_x - 1 \geq N$, and $S_{\tau_x} > x \geq S_{\tau_x - 1}$ gives $x/(\mu + \epsilon) < \tau_x \leq x/(\mu - \epsilon) + 1$. Similarly $\varsigma_x \geq \tau_x - 1 \geq N$ and $S_{\varsigma_x} \leq x$ give $\varsigma_x \leq x/(\mu - \epsilon)$. Divide by x, let $x \to \infty$ and then $\epsilon \to 0$. □

The code-R cost of a silent datum is $-\log$ of a share that is uniform near 0 (the proof of Lemma A.18), so it has all moments, and (i)–(iii) cover every application in this section. The bounded-increment bound of (iii) applies to the race between T_0^{-Z} and T^{ε^*}, whose cost differences are bounded above by the largest cost of T_0^{-Z}; for the race between T^{ε^*} and T_{tr} in code R the increments are unbounded, and only the lower bound and the path limits apply.

Corollary A.21 (The phases of selection, coarse data)**.** *Take a pricing with i.i.d. per-datum costs as in Proposition A.14: exactly Q/rand and R/rand (with ρ stated), or Q/det and R/det as a constant-c_{time} cost model (Proposition A.15). Assume $\Delta \geq 2\varepsilon^*$, $\mathbb{P}[e \neq 0] > 0$, $\varepsilon^* > 0$ and $h_{\mathrm{tr}} > h$. The pool is $\{\emptyset, T_0$ (and $T_0^{-Z}), T^{\varepsilon^*}, T_{\mathrm{tr}}\}$, with $a_\emptyset = 0$, $a_{T_0} = \lambda\ell_0$, $a_{T^{\varepsilon^*}} = \lambda(\ell_0 + h)$ and $a_{T_{\mathrm{tr}}} = \lambda(\ell_0 + h_{\mathrm{tr}})$. Almost surely:*

- *(a) the exact law is refuted for the first time at a geometric index of mean $1/p_\Delta$, and until then T_0 and T_0^{-Z} coincide;*
- *(b) T_{tr} is eventually the unique λ-MAP; the first time it beats T^{ε^*} has mean above $\lambda(h_{\mathrm{tr}} - h)/m_{T^{\varepsilon^*}}$, and at most $(\lambda(h_{\mathrm{tr}} - h) + \gamma)/m_{T^{\varepsilon^*}}$ under code Q, where the costs of T^{ε^*} are bounded by γ;*
- *(c) without T_{tr}, the eventual λ-MAP is the form of least rate among $m_\emptyset$, $m_{T_0^{-}z}$ and $m_{T^{\varepsilon^*}}$ (T^{ε^*} against T_0^{-Z} by Proposition A.16, ties by Theorem A.20(ii));*
- *(d) the expected switch points are: $\emptyset \to T_0$ at $\lambda\ell_0/\gamma$ (deterministic, if T_0 is not yet refuted); $T_0 \to T_0^{-Z}$ at the first refutation, of mean $1/p_\Delta$; $T_0^{-Z} \to T^{\varepsilon^*}$, if $m_{T_0^{-}z} > m_{T^{\varepsilon^*}}$, at a time of mean above $\lambda h/(m_{T_0^{-}z} - m_{T^{\varepsilon^*}})$, and at most $(\lambda h + \gamma_{\mathrm{ex}})/(m_{T_0^{-}z} - m_{T^{\varepsilon^*}})$ when γ_{ex} is bounded; and $T^{\varepsilon^*} \to T_{\mathrm{tr}}$ as in (b).*

When these expected switch points increase in this order, the predicted order of the λ-MAPs is $\emptyset \to T_0 \to T_0^{-Z} \to T^{\varepsilon^*} \to T_{\mathrm{tr}}$.

Proof. Theorem A.20 with the costs of Proposition A.14 and Lemma A.13: refutation by T_0 has probability p_Δ per datum, independently, and while unrefuted T_0 proves every datum (it is never silent), so $J_\lambda(T_0; D_n) = \lambda\ell_0 < n\gamma = J_\lambda(\emptyset; D_n)$ once $n > \lambda\ell_0/\gamma$. T_{tr} has rate 0 and every other form a positive rate ($p_\Delta > 0$, $s_\Delta > 0$, $\gamma_{\mathrm{si}} \geq \log \mathrm{e} - 1 > 0$), which gives (b) with Theorem A.20(iii) at level $\lambda(h_{\mathrm{tr}} - h)$. The bound in (d) for $T_0^{-Z} \to T^{\varepsilon^*}$ is Theorem A.20(iii) applied to the race between these two forms from $n = 0$, at level λh. □

A simulation of the Q/det cost model ($\ell_0 = 200$, $h = 20$, $h_{\mathrm{tr}} = 2000$, $c_{\mathrm{time}} = 22$; 400 runs per setting; code in the accompanying repository) shows the five phases in this order when the predicted switch points are ordered: for a sinusoidal error at $\Delta = 60'$ and $\lambda = 1$ (rates 0.323 and 0.283) the mean switch time to the honest law is 529.8, inside the interval $[495.6, 1251.3]$ given by (d), the mean switch time to the true theory is 7010.1, inside $[6994.9, 7024.9]$, and the mean first refutation is 91.8 against 94.25; when the two rates are close, the patched and the honest law alternate several times before the honest law settles. With rates tuned to be equal, the sign of $J_\lambda(T_0^{-Z}) - J_\lambda(T^{\varepsilon^*})$ changes 148 times on average up to $n = 2 \times 10^5$, and 43% of runs still change sign after $n = 10^5$. Under SAI's pricing with exact theory-machine costs (R/rand, sinusoid, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$; rates 0.1749 and 0.02953), the mean first time the honest law is the λ-MAP is 133.6 against a predicted switch near 138, and the mean first time the true theory is is 67060 against 67055; all 400 runs reach it.

Proposition A.22 (Humility)**.** *Let* $\Pi_\lambda(T \mid D) \propto 2^{-J_\lambda(T;D)}$ *be the power-prior posterior (Hänni and Claude (Anthropic), 2026, Definition 3.6) restricted to a finite pool of deterministic forms (code Q) containing* $\emptyset$, T_0^{-Z}, T^{ε^*} *and possibly* T_{tr}*, and ask a new question at a fresh time* t'*, with cell width* $\Delta' \leq 2\varepsilon^*$.

(a) The pool's three-valued prediction (the analogue, for the pool, of the trichotomy of **??***) puts mass* $\Pi_\lambda(T^{\varepsilon^*} \mid D) + \Pi_\lambda(\emptyset \mid D)$ *on "independent", mass* $\Pi_\lambda(T_0^{-Z} \mid D)$ *on the cell of* $f(t')$*, which is wrong with probability* $p_{\Delta'}$ *over the dither, and mass* $\Pi_\lambda(T_{\mathrm{tr}} \mid D)$ *on the true cell.*

(b) Without T_{tr}*, if the least rate at the training cell width* Δ *is attained by one form only, the posterior concentrates on it almost surely: the inducer ends humble (independent mass tending to* 1 *on finer questions) if* T^{ε^*} *or* $\emptyset$ *is selected, and overconfident (a wrong cell with probability* $p_{\Delta'}$*) if* T_0^{-Z} *is; Proposition A.16 at* Δ *and Proposition A.19 decide which. If two forms tie in rate with a non-degenerate cost difference of finite variance, the posterior oscillates between them forever.*

(c) In a randomised pool (code R) a band-confined completion $\mathrm{BC}_{\mathcal{G}}$ *is not silent: it puts on each cell its* $\mathcal{G}$*-mass, a spread over the cells its band meets and never probability one on a wrong cell; the exact law with random exceptions puts probability* $1 - \rho$ *on the cell of* $f(t')$.

Proof. (a) T^{ε^*} is silent on every question with $\Delta' \leq 2\varepsilon^*$ (Lemma A.13), and $\emptyset$ is silent. T_0^{-Z} proves the cell of $f(t')$, since t' is fresh and so not a patched time, and T_{tr} proves the true cell. (b) Theorem A.20(i), (ii), with $\Pi_\lambda(X \mid D)/\Pi_\lambda(Y \mid D) = 2^{-(J_\lambda(X;D) - J_\lambda(Y;D))}$, whose exponent diverges linearly when the rates differ. (c) The definition of $\mathrm{BC}_{\mathcal{G}}$, and the withdrawal probability ρ. □

body	ε^*, minimax fit (12 / 100 / 2000 yr)	pk, minimax fit	pk, mean fit
Mercury	$0.38'$ / $2.02'$ / $37.4'$	2.80–3.19	3.29–4.71
Venus	$0.36'$ / $0.47'$ / $3.6'$	2.27–3.65	2.81–4.07
Earth–Moon barycentre	$0.27'$ / $0.62'$ / $7.3'$	2.34–3.95	2.91–4.03
Mars	$1.44'$ / $3.04'$ / $49.9'$	2.71–3.38	3.33–4.47
Jupiter	$0.70'$ / $6.57'$ / $34.0'$	1.60–2.65	3.00–7.08
Saturn	$0.17'$ / $7.46'$ / $81.9'$	1.66–2.39	3.21–6.91
Moon (1 month / 1 yr / 18.6 yr)	$30.9'$ / $156'$ / $480'$	1.89–2.21	1.89–3.17

Table 1: Longitude errors of one fixed Kepler ellipse per body and window, against the plan94 and moon98 ephemerides (not the sky): the largest error of the minimax fit, and the peakedness of each fit over the windows of 12 to 2000 years (for the Moon, one month to 18.6 years). Computed; code in the accompanying repository.

The Kepler residuals. Table 1 summarises the error laws. For each body and window, one fixed Kepler ellipse (seven elements) was fitted four ways (three-dimensional least squares, least squares on longitude and latitude, a robust fit for small mean error, and an L_p continuation for small maximum error), and two series of longitude errors at 8000 uniform times were kept: that of the fit with the least sample maximum (the *minimax fit*) and that of the fit with the least sample mean (the *mean fit*). The ephemerides are plan94 for the planets and moon98 for the Moon. An honest law whose tolerance is the sample maximum is slightly false if the window's true supremum is larger. A local parabola through the three time-sorted samples around the maximum estimates the excess at a median 2×10^{-4} (relative) and at most 1.9% in 33 of the 39 series apart from one, Mercury over 2000 years, where the estimate is unreliable because the sampling interval of about 91 days is comparable to Mercury's 88-day period. The band machines therefore use $\varepsilon = (1 + \kappa_\mathrm{b})\varepsilon^*$ with $\kappa_\mathrm{b} \in \{0, 0.01, 0.05\}$, and every verdict below is the same for the three values. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand (and under R/det at $\lambda = 1$ and 4), the uniform band has the least rate in all 429 cells, with exact closed-form costs; in that pool's tolerance family with whole-circle exceptions the optimum is a tolerance of about 0.7–1.0 ε^* with at most 1% of the data as exceptions. With band-confined completions, the histogram machine SH has the least rate in all 429 cells at each κ_b, between 0.4% and 25.7% below the uniform band (median 6.6%), and lies within 0.0011 bits of the Gibbs bound of its test half in every coarse-regime pair; the best two-point form beats the uniform band in 379 of the 429 cells at $\kappa_\mathrm{b} = 0.01$, and in the coarse regime in 34 of the 39 body–window pairs, with best ρ between 0.52 and 0.997, exactly where the criterion of Proposition 1.3(d) says. The five pairs in which the uniform band beats every two-point form are Jupiter over 12 and 30 years, Saturn over 100 years and the Moon over one month and 18.6 years, whose minimax-fit peakedness is 1.60–1.96. The gain persists at fine precision: for the Earth–Moon barycentre over 100 years at $\Delta = 1'$ the rates of the uniform band, the best two-point form and SH are 0.651, 0.560 and 0.503 bits per datum, and for Mars over 12 years 1.660, 1.558 and 1.496. Under Q/det (naming a time costs $c_\mathrm{time} = \log(\text{days in the window}) + 2$ bits, 14.1–21.5 for the planets and 6.8, 10.5 and 14.7 for the Moon's windows; a finite-horizon cost model) the threshold $(\lambda c_\mathrm{time} + \gamma)/(2\gamma)$ grows as the cells coarsen. At $\lambda = 1$ the exact law with exceptions wins from cell widths of about ε^* upwards, the honest law only at the coarsest cells (30°, and from 10° for Jupiter over 30 years and Saturn over 2000 years) and not even there for 17 of the 39 pairs, and at finer cells no law pays; the best member of the tolerance family is then a small tolerance (0.05–0.5 ε^*) with exception shares up to tens of per cent. At $\lambda = 2$ the honest law takes over at cell widths from $8.1'$ to $962'$ (median

$134'$), and at $\lambda = 4$ from $0.72'$ to $962'$ (median $13.4'$). Under Q/rand the exact law with random exceptions wins almost everywhere.

A.5 Derivation horizons and sound reasoning with approximate models

Work over $B = \mathrm{RCF}$ with constants $x_0, x_1, \ldots$ for the states of a system in $\mathbb{R}^m$ (one constant per coordinate), a rational initial state a, a map Ψ definable in B, and a norm $|\cdot|$ for which B proves that Ψ is L-Lipschitz (for a linear $\Psi(x) = Ax$ with rational A and the sup-norm, $L = \|A\|_\infty$, provably in RCF). The *honest dynamical law* is $T^\varepsilon_\Psi := \{|x_0 - a| \le \varepsilon^{(0)}\} \cup \{|x_{s+1} - \Psi(x_s)| \le \varepsilon : s \ge 0\}$, and the *exact law* is $T^0_\Psi := \{x_0 = a\} \cup \{x_{s+1} = \Psi(x_s) : s \ge 0\}$.

Proposition A.23 (Derivation horizon)**.**

(a) For every $L \ge 0$, $T^\varepsilon_\Psi \vdash |x_r - \Psi^r(a)| \le \varepsilon^{(r)}$, with $\varepsilon^{(r)} := L^r\varepsilon^{(0)} + \varepsilon(L^r - 1)/(L - 1)$ for $L \ne 1$ and $\varepsilon^{(r)} := \varepsilon^{(0)} + r\varepsilon$ for $L = 1$.

(b) Let $\Psi(x) = Ax$ be linear, with sup-norm tolerances. The values of the i-th coordinate $x_{r,i}$ over the models of $B \cup T^\varepsilon_\Psi$ form exactly the interval $(A^r a)_i \pm h_{r,i}$, where $h_{r,i} := \varepsilon^{(0)}\|(A^r)_{i\cdot}\|_1 + \varepsilon\sum_{s<r}\|(A^s)_{i\cdot}\|_1$ (the projection of a zonotope). So T^ε_Ψ proves "$x_{r,i} \in [v - \delta, v + \delta]$" iff that interval contains $(A^r a)_i \pm h_{r,i}$. In dimension one, $h_r = \varepsilon^{(r)}$ with $L = |A|$, so (a) is tight.

(c) Put $r^*(\delta) := \max\{r : \varepsilon^{(r)} \le \delta\}$. For $\varepsilon^{(0)} = 0$: $r^*(\delta) = \lfloor \delta/\varepsilon \rfloor$ for $L = 1$; $r^*(\delta) = \lfloor \log_L(1 + \delta(L - 1)/\varepsilon) \rfloor$ for $L > 1$; and for $L < 1$, $r^*(\delta) = \infty$ iff $\delta \ge \varepsilon/(1 - L)$, and $r^*(\delta) = \lfloor \log_L(1 - \delta(1 - L)/\varepsilon) \rfloor$ otherwise. For every L-Lipschitz Ψ, T^ε_Ψ proves a datum at precision δ about x_r, centred at $\Psi^r(a)$, if $r \le r^*(\delta)$; for linear Ψ in dimension one with $L = |A|$, also only if.

Proof. (a) By induction, $|x_{s+1} - \Psi^{s+1}(a)| \le |x_{s+1} - \Psi(x_s)| + |\Psi(x_s) - \Psi(\Psi^s(a))| \le \varepsilon + L\varepsilon^{(s)}$; each step is an instance of the axioms and of the provable Lipschitz inequality, and the recursion $\varepsilon^{(s+1)} = L\varepsilon^{(s)} + \varepsilon$ has the stated solution. (b) $x_r - A^r a = A^r(x_0 - a) + \sum_{s<r} A^{r-1-s}u_s$ with $u_s := x_{s+1} - Ax_s$, and in the real model the deviations $x_0 - a$ and u_s range freely over sup-norm boxes of half-widths $\varepsilon^{(0)}$ and ε. A linear image of a product of boxes is a zonotope, which is convex; its projection on coordinate i is the interval of half-width the sum of the ℓ_1-norms of the corresponding rows, with the extreme values attained by sign choices. So exactly the stated interval of values occurs in $\mathbb{R}$. Since RCF is complete and the claim about $x_{r,i}$ involves only the finitely many axioms with $s < r$, T^ε_Ψ proves an interval statement about $x_{r,i}$ iff it holds in every real model, that is iff the interval contains $(A^r a)_i \pm h_{r,i}$. (c) Solve $\varepsilon^{(r)} \le \delta$; the "if" is (a), and the "only if" for linear Ψ in dimension one is (b). □

For non-linear Ψ the "only if" fails. Let $\Psi(x) = 0$ on $[-1, 1]$ and $\Psi(x) = 2(x - \operatorname{sign} x)$ outside, so that $L = 2$, with $a = 0$, $\varepsilon^{(0)} = 0$ and $\varepsilon = 1/10$. Then $|x_s| \le 1/10 \le 1$ gives $\Psi(x_s) = 0$ and so $|x_{s+1}| \le 1/10$; by induction T^ε_Ψ proves "$x_r \in [-1/10, 1/10]$" at every depth, while $\varepsilon^{(r)} = (2^r - 1)/10$ gives $r^*(1/10) = 1$. A computation (code in the accompanying repository) confirms the closed form against exact interval propagation for $r \le 200$ and several L between 0.5 and 2; at $\delta/\varepsilon = 10^4$ the horizons are 10^4, 463, 72 and 13 for $L = 1$, 1.01, 1.1 and 2, and infinite for $L \le 0.99$. In dimensions 3 and 10, for random matrices of spectral radius 0.95–1.05, propagating boxes by interval arithmetic overestimates the exact tolerance $h_{r,i}$ by a median factor of 5–300 at $r = 10$ and up to 10^9 at $r = 30$, and the ∞-norm Lipschitz bound by up to 10^{13}: a reasoner that propagates boxes has a much shorter effective horizon than the theory.

Proposition A.24 (The horizon of an exact approximation)**.** *Suppose the true trajectory satisfies T^{ε}_{Ψ}, but the inducer holds the exact law T^{0}_{Ψ} with the same a. Then T^{0}_{Ψ} is consistent and proves $x_r = \Psi^r(a)$, which is within $\varepsilon^{(r)}$ of the true x_r. If $r \le r^*(\delta)$, the depth-r conclusions of T^{0}_{Ψ} are δ-correct for every trajectory compatible with the honest law. For linear Ψ in dimension one with $L = |A|$, some trajectory compatible with the honest law attains the deviation $\varepsilon^{(r)}$, so in that case they are δ-correct for all such trajectories only if $r \le r^*(\delta)$. And T^{0}_{Ψ} together with one true datum about x_r whose interval excludes $\Psi^r(a)$ is inconsistent, and so proves every sentence.*

Proof. The exact trajectory is a model of T^{0}_{Ψ}, and the true trajectory is a model of T^{ε}_{Ψ}. The middle claims are Proposition A.23(a), (b). For the last, the datum and $x_r = \Psi^r(a)$ are jointly refutable in B. □

Proposition A.25 (Sound reasoning with approximate models)**.** *Let $T_i := \{\forall t \in \mathrm{Dom}_i : \ |q(t) - f_i(t)| \le \varepsilon_i\}$, $i \le k$, be honest pointwise models of one quantity q (scalar, or vector-valued with a norm), with q otherwise unconstrained, and let T^{0}_{i} be their exact versions ($\varepsilon_i = 0$).*

(a) Let B be true in the intended structure. If every T_i is true, then $\bigcup_i T_i$ is consistent and every sentence derivable from $B \cup \bigcup_i T_i$ is true; every derived statement about $q(t)$ is a set containing the true value.

For (b)–(d) assume moreover that B decides every sentence built from the f_i, the Dom_i and the ε_i at the times concerned (for example $B = \mathrm{RCF}$, with semialgebraic f_i and Dom_i and rational ε_i).

(b) $\bigcup_i T^{0}_{i}$ is inconsistent, and so proves every sentence, iff $f_i(t) \ne f_j(t)$ for some $t \in \mathrm{Dom}_i \cap \mathrm{Dom}_j$.

(c) $\bigcup_i T_i$ is consistent iff for each t the balls $\{v : |v - f_i(t)| \le \varepsilon_i\}$ with $t \in \mathrm{Dom}_i$ have a common point. For scalar q, and for vector q with sup-norm (box) tolerances, this holds iff it holds pairwise, $|f_i(t) - f_j(t)| \le \varepsilon_i + \varepsilon_j$ on every overlap. For Euclidean tolerances in $\mathbb{R}^m$, $(m+1)$-wise intersection is necessary and sufficient, and pairwise intersection is not sufficient.

(d) If B proves that a map Ψ is L_Ψ-Lipschitz, then $T_i \vdash |\Psi(q(t)) - \Psi(f_i(t))| \le L_\Psi \varepsilon_i$.

Proof. (a) The intended structure is a model, and derivations preserve truth. (b) If two exact models disagree at a common time t, together they prove $f_i(t) = f_j(t)$, which B refutes by the hypothesis. If they agree on all overlaps, then in every model of B (in which B's decisions fix the values $f_i(t)$) setting $q(t) := f_i(t)$ for $t \in \mathrm{Dom}_i$, and arbitrarily elsewhere, gives a model. (c) The constants $q(t)$ for distinct t are distinct and otherwise unconstrained, so by compactness $B \cup \bigcup_i T_i$ is consistent iff for each t, B is consistent with "some v lies in all the balls at t". By the hypothesis B decides this sentence, and B is true in the intended structure, so it is consistent with B iff it holds there, that is iff the balls have a common point. Intervals that meet pairwise have $\max_i$(left end) $\le \min_j$(right end), which is a common point (Helly's theorem in dimension one), and boxes are products of intervals. Helly's theorem in $\mathbb{R}^m$ (standard; Helly, 1923) gives the $(m+1)$-wise criterion. Three discs of radius 1.1 centred at the vertices of an equilateral triangle of side 2 meet pairwise but have no common point, since the circumradius is $2/\sqrt{3} \approx 1.155$. (d) is the Lipschitz inequality in B. □

Without the hypothesis on B, (b) and (c) fail: take $B = \emptyset$, $f_i(t)$ and $f_j(t)$ distinct constant symbols and $\mathrm{Dom}_i = \mathrm{Dom}_j$ a single time; then $T^{0}_{i} \cup T^{0}_{j}$ is consistent (interpret the two constants as equal), although the values differ in the intended structure. A computation finds pairwise and

joint overlap in agreement on 10^5 random families of 2 to 6 intervals and on 2×10^4 random families of boxes in $\mathbb{R}^2$.

A.6 Best systems

Proposition A.26 (Marginal slopes)**.** *Let T^* be a λ-MAP within a family of theories that contains $T^* \setminus \mathcal{A}$ and $T^* \cup \mathcal{B}$ for the modules considered. For every module $\mathcal{A} \subseteq T^*$, $\mathrm{Exc}_D(T^* \setminus \mathcal{A}) - \mathrm{Exc}_D(T^*) \geq \lambda \|\mathcal{A}\|$; and for every $\mathcal{B}$ disjoint from T^*, $\mathrm{Exc}_D(T^*) - \mathrm{Exc}_D(T^* \cup \mathcal{B}) \leq \lambda \|\mathcal{B}\|$.*

Proof. $J_\lambda(T^*; D) \leq J_\lambda(T^* \setminus \mathcal{A}; D)$ and $J_\lambda(T^*; D) \leq J_\lambda(T^* \cup \mathcal{B}; D)$, with $\|T^*\| - \|T^* \setminus \mathcal{A}\| = \|\mathcal{A}\|$ and $\|T^* \cup \mathcal{B}\| - \|T^*\| = \|\mathcal{B}\|$. □

Proof of Proposition 1.5. By the hypotheses (Proposition A.7) the hull lemma (Hänni and Claude (Anthropic), 2026, Lemma 3.8(v)) applies: for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are exactly the theories at v_i, and at $\lambda = \sigma_j$ ($j \geq 1$) they are the theories on the closed edge $[v_{j-1}, v_j]$. (b)⇒(a): $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$. (a)⇒(b): if some edge slope σ_j lay in (τ, σ), the λ-MAP set at $\lambda = \sigma_j$, the theories on $[v_{j-1}, v_j]$, would differ from the set just below σ_j, the theories at v_j only, since $v_{j-1} \neq v_j$. So no edge slope lies in (τ, σ), and (τ, σ) lies in one interval (σ_{i+1}, σ_i), which is (b). (b)⇔(c) is Hänni and Claude (Anthropic) (2026, Corollary 5.4(c)): a zero-window (σ, τ)-kink with $\tau < \sigma$ is a vertex v_i with $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$, and conversely every vertex v_i is a zero-window (σ_i, σ_{i+1})-kink, hence a (σ, τ)-kink whenever $\sigma \leq \sigma_i$ and $\tau \geq \sigma_{i+1}$. □

A computation on 400 random point families with 10 random intervals each, testing "the λ-MAP set is constant on (τ, σ)" exactly at every crossing point and between consecutive ones, agrees with (b) in all 4000 cases.

Proposition A.27 (The trivialisation is blocked)**.** *Let $\mathcal{L}' \supseteq \mathcal{L}$ be any extension of the language, for example by a predicate* Act *true of exactly the actual things, with the axiom $\forall x\, \mathrm{Act}(x)$; fix a syntax code for $\mathcal{L}'$, and let the data be base-language answers whose provability from the theories considered is decidable. Then every finite theory T in $\mathcal{L}'$ consistent with D has $\|T\| + \mathrm{Exc}^{\mathcal{Q}}_D(T) \geq K(y_D \mid \mathrm{dom}\, D) - K(\mathcal{L}') - c$, where $K(\mathcal{L}')$ is the complexity of a decoder for $\mathcal{L}'$ and its syntax code. Without decidability, add n on the left. If $\mathcal{L}'$ and its code are fixed independently of D, $K(\mathcal{L}')$ is a constant.*

Proof. The weights in the proof of Proposition A.8 search for, or decide, proofs from T of base-language answers. Given a decoder for the syntax of $\mathcal{L}'$ they are computable (respectively lower semicomputable) from T and the questions, and they do not otherwise depend on the language. So the argument of Proposition A.8(b) (respectively (a)) gives $K(y_D \mid \mathrm{dom}\, D) \leq K(\mathcal{L}') + K(T \mid \mathcal{L}') + \mathrm{Exc}^{\mathcal{Q}}_D(T) + c$ (respectively with n added), and $K(T \mid \mathcal{L}') \leq \|T\| + c$. □

Proposition A.28 (Facts used once never enter for $\lambda > 1$)**.**

(a) Let T^* be a λ-MAP with $\lambda > 1$ and χ a sentence. Suppose that, in the context $T^* \setminus \{\chi\}$, adding χ turns exactly the exceptions in a set $\mathcal{D}_\chi \subseteq D$ into proved data and changes nothing else, and that $\sum_{(\mathcal{Q},a) \in \mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda \ell(\chi)$. Then $\chi \notin T^*$. In particular this applies to a datum's answer a stated as an axiom whenever $\gamma(\mathcal{Q}) \leq \ell(a)$ and stating it proves no other datum in that context.

*(b) Let provability on the data be decidable in the class, with κ the constant of Proposition A.8, and let T_s be b-*sufficient*: $\|T_s\| + \mathrm{Exc}_D(T_s) \leq K(y_D \mid \mathrm{dom}\, D) + b$. Every theory T has $\mathrm{Exc}_D(T) \geq \mathrm{Exc}_D(T_s) - (\|T\| - \|T_s\|) - (b + \kappa)$; a module $\mathcal{A}$ added to T_s has $\mathrm{EP}_D(\mathcal{A}; T_s \cup \mathcal{A}) \leq 1 + (b + \kappa)/\|\mathcal{A}\|$; and every λ-MAP T with $\lambda > 1$ has $\|T\| \leq \|T_s\| + (b + \kappa)/(\lambda - 1)$.*

Proof. (a) If $\chi \in T^*$, deleting it raises Exc_D by exactly $\sum_{\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi) = \lambda\|\{\chi\}\|$, contradicting Proposition A.26. For an answer stated as an axiom, $\mathcal{D}_\chi$ is the one datum, and $\gamma(\mathcal{Q}) \le \ell(a) < \lambda\ell(a)$. (b) Proposition A.8(b) gives $\|T\| + \mathrm{Exc}_D(T) \ge K(y_D \mid \mathrm{dom}\, D) - \kappa \ge \|T_s\| + \mathrm{Exc}_D(T_s) - b - \kappa$, which is the first claim, a right line of slope 1 through the point of T_s. For a module $\mathcal{A}$ disjoint from T_s, $\|T_s \cup \mathcal{A}\| - \|T_s\| = \|\mathcal{A}\|$, which gives the second. For a λ-MAP T, $\lambda\|T\| + \mathrm{Exc}_D(T) \le \lambda\|T_s\| + \mathrm{Exc}_D(T_s)$; combined with the first claim this gives $(\lambda-1)(\|T\| - \|T_s\|) \le b + \kappa$, which is Hänni and Claude (Anthropic) (2026, Corollary 5.5(a)) in this class. □

So at $\lambda = 1$ a fact that explains only itself is excluded only by the code's per-axiom overhead, while every $\lambda > 1$ excludes every module whose explanatory power is below λ.

The solar-system toy. Bodies $b \le k_\mathrm{b}$ have n_b observations each, each of literal cost γ (code Q), and $n_\mathrm{tot} := \sum_b n_b$. The modules are the law Λ (L bits), body b's elements El_b (k bits), and Kepler's third law K3 (L_3 bits), which shortens every elements module by s bits, $0 < s < k$, because it derives the period from the semi-major axis. $\Lambda \cup \mathrm{El}_b$ proves all of b's observations, and no other combination proves any observation: laws alone answer no position question, and neither do elements alone. The theories are Λ or not, K3 or not, and a set S of bodies whose elements are axioms.

Proposition A.29 (Laws and initial conditions in the solar-system toy)**.** *Put $k_\Lambda := k$, $k_{\Lambda+\mathrm{K3}} := k - s$ and $S_x(\lambda) := \{b : \gamma n_b > \lambda k_x\}$ for $x \in \{\Lambda, \Lambda + \mathrm{K3}\}$. The optimal values of the three law choices are*

$$J^*_\emptyset = \gamma n_\mathrm{tot}, \qquad J^*_\Lambda = \lambda L + \sum_b \min(\gamma n_b, \lambda k), \qquad J^*_{\Lambda+\mathrm{K3}} = \lambda(L + L_3) + \sum_b \min(\gamma n_b, \lambda(k-s)),$$

*attained by $S = S_x(\lambda)$ (bodies with $\gamma n_b = \lambda k_x$ may go either way); the λ-MAP is the best of the three (*K3 *without Λ is useless).*

(a) Λ beats the empty theory iff $\sum_{b \in S_\Lambda(\lambda)} (\gamma n_b - \lambda k) > \lambda L$.

(b) $J^*_{\Lambda+\mathrm{K3}} - J^*_\Lambda = \lambda L_3 - \sum_b \big[\min(\gamma n_b, \lambda k) - \min(\gamma n_b, \lambda(k-s))\big]$. If no body has $\lambda(k-s) < \gamma n_b \le \lambda k$, so that S is the same with and without K3*, then $J^*_{\Lambda+\mathrm{K3}} < J^*_\Lambda$ iff $s|S| > L_3$, whatever λ is.* K3 *is in the λ-MAP iff in addition $J^*_{\Lambda+\mathrm{K3}} < J^*_\emptyset$.*

(c) In the λ-MAP $\Lambda \cup \{\mathrm{El}_b : b \in S\}$ (with or without K3*), $\mathrm{EP}(\Lambda) = \gamma \sum_{b \in S} n_b / L$ and $\mathrm{EP}(\mathrm{El}_b) = \gamma n_b / k_x$.*

Proof. Without Λ no observation is proved, so the best such theory is empty, with $J = \gamma n_\mathrm{tot}$. For a law choice x with Λ, J_λ is a constant plus a sum over bodies, where including El_b costs λk_x and excluding it costs γn_b; minimise each term. (a) is the comparison of J^*_Λ with $J^*_\emptyset$. (b) The bracket is λs when $\gamma n_b > \lambda k$, $\gamma n_b - \lambda(k-s)$ when $\lambda(k-s) < \gamma n_b \le \lambda k$, and 0 otherwise; without bodies in the middle range the difference is $\lambda(L_3 - s|S|)$. The λ-MAP is the best of the three law choices, the empty one included. (c) Deleting Λ (respectively El_b) makes exactly the observations of the bodies in S (respectively of b) literal. □

Part (b) compares law choices; it is not membership of the λ-MAP. With two bodies of one observation each, $\gamma = 10$, $L = 1000$, $k = 5$, $s = 4$, $L_3 = 1$ and $\lambda = 1$, S is both bodies with and without K3 and $s|S| = 8 > L_3$, but the λ-MAP is the empty theory: $J^*_\emptyset = 20$ against $J^*_{\Lambda+\mathrm{K3}} = 1003$ and $J^*_\Lambda = 1010$. The middle-range condition of (b) matters too: in the population

of Section 1.5 (eight planets with 655 to 3838 observations, sixty asteroids with a median of about five, thirty comets with one or two; $L = 300$, $L_3 = 150$, $k = 60$, $s = 15$, $\gamma = 12.4$), K3 is in the λ-MAP at $\lambda = 6.75$ although $s|S_\Lambda(\lambda)| = 135 < L_3$, because with K3 more bodies pay for their elements. A computation (code in the accompanying repository) confirms the closed form against brute force over all theories in 2100 random cases, and the criterion of (b) in all 13250 random cases with the same S under both law choices.

Proposition A.30 (Laws and initial conditions in the question format)**.** *Let the data be conditional questions "given the elements α_b of body b, is ℓ the case?", with compatibility as in* (E3)*: a datum is compatible with T when $B \cup T \cup \{\alpha_b\}$ is consistent and proves ℓ. In the toy, every theory of the family (the law,* K3*, and elements modules stating the true elements) is consistent with every true antecedent. Then an elements module* El_b *with the same content as α_b adds nothing, so* $\mathrm{EP}(\mathrm{El}_b) = 0$ *and no λ-MAP contains elements; and Λ alone answers every question, with* $\mathrm{EP}(\Lambda) = \gamma n_{\mathrm{tot}}/L$ *however the n_{tot} observations are spread over bodies.*

Proof. Immediate from the definition of compatibility: $T \cup \{\alpha_b\} \cup \mathrm{El}_b$ and $T \cup \{\alpha_b\}$ prove the same sentences, and $\Lambda \cup \{\alpha_b\}$ proves all of b's observations. Adding El_b raises J_λ by $\lambda k > 0$ and lowers no exception cost. □

Without the consistency condition the encoding is degenerate: a theory that refutes the antecedents would prove every datum (**??**). In the question format the distinction between laws and initial conditions is the distinction between theory and question antecedent.

A.7 Explanatory power, unification and correction

Proposition A.31 (Explanatory power as a likelihood ratio; unification)**.** *Write $P_T(D) := 2^{-\mathrm{Exc}_D(T)}$ (code Q or R). For a hypothesis h, a background theory $T_\flat$, data sets D_1, D_2 on disjoint questions and a theory x, put $I(h; D \mid T_\flat) := \log(P_{T_\flat \cup h}(D)/P_{T_\flat}(D))$, $I_x(D_1; D_2) := \log \frac{P_x(D_1 \cup D_2)}{P_x(D_1)P_x(D_2)}$ and $U(D_1, D_2; h \mid T_\flat) := I(h; D_1 \cup D_2 \mid T_\flat) - I(h; D_1 \mid T_\flat) - I(h; D_2 \mid T_\flat)$.*

(a) $\|\mathcal{A}\| \,\mathrm{EP}_D(\mathcal{A}; T) = I(\mathcal{A}; D \mid T \setminus \mathcal{A})$.

(b) $U(D_1, D_2; h \mid T_\flat) = I_{T_\flat \cup h}(D_1; D_2) - I_{T_\flat}(D_1; D_2)$*. If P is a product over questions for both theories, then $U = 0$ for every split.*

(c) Let h be a law family with a k-bit parameter θ under the uniform prior, $P_h(D) := 2^{-k} \sum_\theta P_{h \cup \{\theta\}}(D)$, and $P_\emptyset$ a product. If each of D_1, D_2 is proved by $h \cup \{\theta^*\}$ and has likelihood 0 under every other θ, then $U(D_1, D_2; h) = k$. If $\theta = (\theta_1, \theta_2)$ with D_i depending only on θ_i, then $U = 0$. If $\theta_i = (\theta_{\mathrm{sh}}, \theta_i')$ share a k_{sh}-bit constant θ_{sh} and each D_i identifies $(\theta_{\mathrm{sh}}, \theta_i')$, then $U = k_{\mathrm{sh}}$.

Proof. (a) is the definition of EP. (b) Expand the six logarithms: the first term of the difference is $\log P_{T_\flat \cup h}(D_1 \cup D_2) - \log P_{T_\flat \cup h}(D_1) - \log P_{T_\flat \cup h}(D_2)$, and the second is the same expression for $T_\flat$; a product law makes $I_x(D_1; D_2) = 0$. (c) Use (b) with $I_\emptyset = 0$. In the first case $P_h(D_1) = P_h(D_2) = P_h(D_1 \cup D_2) = 2^{-k}$, so $I_h(D_1; D_2) = k$. In the second, $P_h(D_1 \cup D_2) = P_h(D_1)P_h(D_2)$. In the third, with θ_i' of k' bits, $P_h(D_i) = 2^{-k_{\mathrm{sh}} - k'}$ and $P_h(D_1 \cup D_2) = 2^{-k_{\mathrm{sh}} - 2k'}$, so $I_h(D_1; D_2) = k_{\mathrm{sh}}$. □

A computation with $k = 6$ and observations given by a random table gives $U = 0.00$ for a stated parameter, 6.00 for a marginalised parameter and one body, 0.00 for two bodies with independent parameters, and 3.00 for two bodies sharing a 3-bit constant; the identity (b) holds to 3×10^{-15} on 1000 random joint laws. Myrvold describes unification as the degree to which a hypothesis

renders phenomena informationally relevant to one another, a contribution to evidential support (Myrvold, 2003); by (b), U is the change in the mutual information between D_1 and D_2 that adding h brings about, which has that form. We do not claim that his formula coincides with U in every detail.

Proposition A.32 (Unification without better prediction)**.** *Consider three theories with a common part:* Pt_f*, Ptolemy's system with free epicycle phases, which pays for the coincidence of each superior planet p's retrograde motion with its opposition in its fit,* a_p *bits per planet;* Pt_s*, the same with the coincidences stated as axioms,* $\sum_p a_p$ *law bits more than* Pt_f *and with the fit of* Cop*; and* Cop*, Copernicus's system,* H *law bits more than* Pt_f *and with the fit of* Pt_s*.*

(a) Cop *beats* Pt_s*, which makes the same predictions, at every* λ *iff* $H < \sum_p a_p$*.*

(b) Cop *is* Pt_f *plus a module with* $\mathrm{EP} = \sum_p a_p/H$*, so it beats* Pt_f *iff* $\lambda < \sum_p a_p/H$*.*

(c) Pt_s *beats* Pt_f *iff* $\lambda < 1$*.*

Hence if $H < \sum_p a_p$*,* Cop *is the best of the three exactly for* $\lambda < \sum_p a_p/H$*, and* Pt_f *above; if* $H \geq \sum_p a_p$*,* Cop *is never strictly best:* Pt_s *is best for* $\lambda < 1$ *(tying* Cop *if* $H = \sum_p a_p$*) and* Pt_f *for* $\lambda > 1$*.*

Proof. Measure lengths and fits relative to Pt_f and write $a := \sum_p a_p$. Then $J_\lambda(\mathrm{Pt}_f) = 0$, $J_\lambda(\mathrm{Pt}_s) = \lambda a - a$ and $J_\lambda(\mathrm{Cop}) = \lambda H - a$. (a) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_s) = \lambda(H - a)$, negative for every $\lambda > 0$ iff $H < a$. (b) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_f) = \lambda H - a$, and the module of H bits lowers the fit by a. (c) $J_\lambda(\mathrm{Pt}_s) - J_\lambda(\mathrm{Pt}_f) = (\lambda - 1)a$. If $H < a$, then $a/H > 1$: for $\lambda < a/H$, Cop beats both others; for $\lambda > a/H > 1$, Pt_f beats both. If $H \geq a$: for $\lambda < 1$, Pt_s beats Pt_f and is at least as good as Cop, with equality iff $H = a$; for $\lambda > 1 \geq a/H$, Pt_f beats both; at $\lambda = 1$, Pt_s and Pt_f tie and Cop is no better. □

Newton's theory does not entail Kepler's laws: it contradicts them (cf. Duhem, 1954). What it derives, with masses and initial conditions, are corrected statements, and Kepler's laws within tolerances on windows, as in the model of reduction to a corrected reduced theory (cf. Schaffner, 1967). The following toy makes this provable. Over $B = \mathrm{RCF}$ with constants $x_{t,1}, x_{t,2}$ for $t \leq \bar{t}$ (a planar state x_t per step), let Rot be the rotation with cosine $3/5$ and sine $4/5$, $\mathbf{P}$ a rational 2×2 matrix with operator norm $\|\mathbf{P}\|$, $\eta_c := 1/100$, $A := \mathrm{Rot} + \eta_c \mathbf{P}$ and $a := (1, 0)$; $|\cdot|$ is the Euclidean norm, and $|v| \leq \varepsilon$ is stated as $|v|^2 \leq \varepsilon^2$ with ε^2 rational. The *old exact law* is $T_0 := \{x_t = \mathrm{Rot}^t a : t \leq \bar{t}\}$ (uniform circular motion), the *successor* is $T_{\mathrm{new}} := \{x_0 = a\} \cup \{x_{t+1} = Ax_t : t < \bar{t}\}$, the *honest old law* is $T^\varepsilon := \{|x_t - \mathrm{Rot}^t a| \leq \varepsilon : t \leq \bar{t}\}$, and a *time-dependent form* is $T^{(\varepsilon_\cdot)} := \{|x_t - \mathrm{Rot}^t a| \leq \varepsilon_t : t \leq \bar{t}\}$.

Proposition A.33 (Explanation by approximate correction)**.**

(a) $T_{\mathrm{new}} \vdash T^\varepsilon$ *for every rational* $\varepsilon \geq \varepsilon(\bar{t}) := ((1 + \eta_c \|\mathbf{P}\|)^{\bar{t}} - 1)|a|$*.*

(b) $T_{\mathrm{new}} \vdash T^\varepsilon$ *iff* $\varepsilon^2 \geq \varepsilon_{\min}(\bar{t})^2 := \max_{t \leq \bar{t}} |A^t a - \mathrm{Rot}^t a|^2$*, a rational number. So* $\varepsilon_{\min}(\bar{t}) \leq \varepsilon(\bar{t})$*, and* $T^{\varepsilon_{\min}(\bar{t})}$ *is the strongest uniform-tolerance form of the old law on the window that the successor proves.*

(c) If $\eta_c \mathbf{P} a \neq 0$ *and* $\bar{t} \geq 1$*, then* $T_{\mathrm{new}} \vdash \neg \bigwedge T_0$*.*

(d) Among uniform-tolerance forms, T^ε *is a common consequence of* T_0 *and* T_{new} *iff* $\varepsilon \geq \varepsilon_{\min}(\bar{t})$*; the strongest common one is* $T^{\varepsilon_{\min}(\bar{t})}$*.*

(e) The time-dependent form with $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$ *is a common consequence of both theories, strictly stronger than* $T^{\varepsilon_{\min}(\bar{t})}$ *whenever* $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ *for some* t*.*

(f) The strongest common consequence of the two finite theories is their disjunction $\bigwedge T_0 \vee \bigwedge T_{\text{new}}$*, which is not a tolerance law; if* $\varepsilon_{\min}(\bar{t}) > 0$*,* $T^{\varepsilon_{\min}(\bar{t})}$ *does not entail it.*

Proof. Write $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$; ε_t^2 is rational because $A^t a$ and $\mathrm{Rot}^t a$ are rational vectors. In every model of $B \cup T_{\text{new}}$, that is in any real closed field with interpretations of the constants, the recurrence determines $x_t = A^t a$, a rational vector, since $\mathbb{Q}$ embeds uniquely into every ordered field; instantiating the recurrence t times gives $T_{\text{new}} \vdash x_t = A^t a$.

(a) $\|\mathrm{Rot}\| = 1$ and $\|A\| \le 1 + \eta_c \|\mathbf{P}\|$, so $\varepsilon_{t+1} = |A(A^t a - \mathrm{Rot}^t a) + \eta_c \mathbf{P}\,\mathrm{Rot}^t a| \le (1 + \eta_c\|\mathbf{P}\|)\varepsilon_t + \eta_c\|\mathbf{P}\||a|$ with $\varepsilon_0 = 0$, and by induction $\varepsilon_t \le ((1 + \eta_c\|\mathbf{P}\|)^t - 1)|a| \le \varepsilon(\bar{t})$ for $t \le \bar{t}$. Since T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, it proves each conjunct of T^ε for rational $\varepsilon \ge \varepsilon(\bar{t})$.

(b) Over $B \cup T_{\text{new}}$ each conjunct of T^ε is equivalent to the inequality $\varepsilon_t^2 \le \varepsilon^2$ between rationals, which RCF proves or refutes. So $T_{\text{new}} \vdash T^\varepsilon$ iff all $\bar{t} + 1$ inequalities hold, iff $\varepsilon^2 \ge \max_{t \le \bar{t}} \varepsilon_t^2 = \varepsilon_{\min}(\bar{t})^2$; and $\varepsilon_{\min}(\bar{t}) \le \varepsilon(\bar{t})$ by the bound in (a).

(c) $T_{\text{new}} \vdash x_1 = Aa$ and $T_0 \ni (x_1 = \mathrm{Rot}\, a)$, with $Aa - \mathrm{Rot}\, a = \eta_c \mathbf{P} a \neq 0$.

(d) T_0 proves $x_t = \mathrm{Rot}^t a$, hence T^ε for every $\varepsilon \ge 0$; combine with (b).

(e) T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, and T_0 proves $|x_t - \mathrm{Rot}^t a|^2 = 0 \le \varepsilon_t^2$. If $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ at some t, choose a real point x_t with $\varepsilon_t < |x_t - \mathrm{Rot}^t a| \le \varepsilon_{\min}(\bar{t})$ and put $x_s := \mathrm{Rot}^s a$ for $s \neq t$: this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ that violates the time-dependent form.

(f) For finite theories, a sentence follows from both iff it follows from their disjunction (proof by cases), so the disjunction is the strongest common consequence. Put $x_t := \mathrm{Rot}^t a + (\varepsilon_{\min}(\bar{t}), 0)$ for every $t \le \bar{t}$: if $\varepsilon_{\min}(\bar{t}) > 0$, this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ with $x_0 \neq a$, so it is a model of neither theory, and $T^{\varepsilon_{\min}(\bar{t})}$ does not entail the disjunction; in particular the disjunction is not equivalent to a uniform-tolerance law. □

Remark A.34 (Derivations of polynomial size)*. Proof sketch.* What is not written out is the size of derivations of arithmetic on binary numerals from the field axioms, which is standard. The natural derivation of T^ε from T_{new} instantiates the recurrence t times, writes each x_t as an exact rational vector and checks $\bar{t} + 1$ numerical inequalities; the numerals it writes have total bit size $\sum_{t \le \bar{t}} \mathrm{bits}(A^t a) = O(\bar{t}^2 \log \mathrm{ht})$, where ht is the height of the entries of A. If the numeral arithmetic has derivations of size polynomial in the numerals' size, the whole derivation has size polynomial in $\bar{t}$. For $\mathbf{P} = \mathbf{J}\,\mathrm{Rot}$, with $\mathbf{J}$ the rotation by 90°, the numerals have 1894 bits at $\bar{t} = 10$ and about 1.8×10^5 at $\bar{t} = 100$.

Exact rational arithmetic (code in the accompanying repository) gives, for three perturbations with $\|\mathbf{P}\| = 1$ ($\mathbf{J}$ is the rotation by 90°):

$\mathbf{P}$	$\varepsilon_{\min}(\bar{t})$ at $\bar{t} = 1, 10, 100, 200$	bound $\varepsilon(\bar{t})$
$\mathbf{J}\,\mathrm{Rot}$ (outward spiral with precession)	0.0100, 0.0999, 0.961, 1.69	0.0100, 0.105, 1.70, 6.32
$\mathrm{diag}(1, -1)\,\mathrm{Rot}$ (distortion)	0.0100, 0.0127, 0.0186, 0.0243	the same
identity (spiral)	0.0100, 0.103, 1.33, 3.49	the same

In each case T_{new} refutes T_0 at $t = 1$. The bound $\varepsilon(\bar{t})$ is nearly tight for small windows when the perturbation accumulates, and very loose for a bounded distortion (ratio 0.004 at $\bar{t} = 200$). For $\mathbf{P} = \mathbf{J}\,\mathrm{Rot}$ the time-dependent tolerances have mean 0.0500 at $\bar{t} = 10$ and 0.490 at $\bar{t} = 100$, about half of $\varepsilon_{\min}(\bar{t})$. A circle refitted on the window (radius and angular speed; floating point, for comparison only) absorbs the precession almost entirely in the first case (2.5×10^{-4} and 2.5×10^{-3} at $\bar{t} = 10, 100$) and the growth not at all in the third (3.7×10^{-2} and 0.80): the toy analogue of the fitted ellipses of Table 1.

A.8 Domains of applicability and retention

Let a law Λ with per-system elements prove the data of every system but one, $\star$, whose $n_\star$ data it fits, with $\star$'s elements ($k_\star$ bits), at the per-datum rate $m^\star_{\text{pat}}$ of the exact law with exceptions, or at the rate $m^\star_{\text{hon}}$ of an honest tolerance for $\star$, at a one-off cost of $h_\star$ bits. The *domain-restricted law* $\Lambda_{\text{Dom}} := \forall b\,(\text{Dom}(b) \to \text{law}(b))$, with Dom a definable predicate that B proves of the other systems and refutes of $\star$, costs $\|\text{Dom}\|$ bits more than Λ (a few bits if "planet" is a base predicate, about the logarithm of the number of systems to name $\star$) and leaves $\star$'s data literal, at rate γ. So $\star$'s data have three treatments: restriction (rate γ, constant $\lambda\|\text{Dom}\|$), patched (rate $m^\star_{\text{pat}}$, constant $\lambda k_\star$) and honest (rate $m^\star_{\text{hon}}$, constant $\lambda(k_\star + h_\star)$).

Proposition A.35 (Domain restriction)**.**

(a) $J_\lambda(\Lambda_{\text{Dom}}) - J_\lambda(\Lambda \cup \text{El}_\star$ *with* $\star$ *'s exceptions*$) = \lambda\|\text{Dom}\| - \lambda k_\star + \sum_{\star\text{'s data}}(\gamma - c_{\text{pat}})$*, and likewise against the honest treatment of* $\star$*, with* $\lambda(k_\star + h_\star)$ *and* c_{hon}*, where* $c_{\text{pat}}, c_{\text{hon}}$ *are the per-datum costs.*

(b) With i.i.d. data and $n_\star \to \infty$*, the restriction is eventually the* λ*-MAP's treatment of* $\star$ *iff* $\gamma < \min(m^\star_{\text{pat}}, m^\star_{\text{hon}})$*, or* $\gamma = m^\star_{\text{hon}} < m^\star_{\text{pat}}$ *with the honest cost almost surely* γ *and* $\|\text{Dom}\| < k_\star + h_\star$*. If the least rate is shared by two treatments whose costs differ non-degenerately, there is no eventual winner.*

(c) Under SAI's pricing (code R), $m^\star_{\text{hon}} < \gamma$ *whenever* $\varepsilon^*_\star < \mathsf{R}/(2\mathrm{e})$ *(for* $\Delta \le 2\varepsilon^*_\star$*) or* $\Delta < \mathsf{R}/\mathrm{e}$ *(for* $\Delta \ge 2\varepsilon^*_\star$*). So a system that the law confines within about* 66° *of longitude (a sufficient bound) is never cut out asymptotically at any cell width below* R/e*: it is kept with its own, wider tolerance. Under code Q with* $\Delta \le 2\varepsilon^*_\star$ *the honest treatment costs exactly* γ *per datum, and the restriction is eventually selected iff* $m^\star_{\text{pat}} > \gamma$ *and* $\|\text{Dom}\| < k_\star + h_\star$*.*

Proof. (a) The two theories differ in the domain clause, in $\star$'s elements, and in the treatment of $\star$'s data; the other systems' data are proved by both. (b) Theorem A.20(i), (ii) applied to the three treatments (code-Q costs are bounded and code-R costs have all moments): a unique least rate wins eventually; a tie with an almost surely zero cost difference is decided by the constants, which is the case of the restriction and the honest treatment under code Q in the fine regime, where the honest law is silent on every datum and costs exactly γ; and a tie with a non-degenerate difference has no eventual winner. (c) In the fine regime Lemma A.18(ii) gives $m^\star_{\text{hon}} \le \log(2\varepsilon^*_\star/\Delta) + \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $2\mathrm{e}\varepsilon^*_\star < \mathsf{R}$. In the coarse regime $m^\star_{\text{hon}} = s_\Delta \beta_e \le \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $\Delta < \mathsf{R}/\mathrm{e}$. Under code Q with $\Delta \le 2\varepsilon^*_\star$ the honest law is silent on every datum (Lemma A.13), and (b) applies. □

For the Moon over one year, with one fixed ellipse ($\varepsilon^* = 156.3' = 2.6^\circ$, $\mathbb{E}|e| = 61.7'$ for the mean fit, $k_\star = 60$, $h_\star = 20$, $\|\text{Dom}\| = 4$, $c_{\text{time}} = 10.5$), an exact computation (code in the accompanying repository) gives the eventually selected treatment of the Moon's data, with rates in bits per datum ("excluded" is the domain restriction, chosen at an exact tie between the honest and the literal treatment):

Δ	γ	code Q, $\lambda = 1$	code Q, $\lambda = 2$	code R, $\lambda = 1, 2$
$10'$	11.08	excluded (tie; patched 20.52)	excluded (patched 30.51)	honest (4.973)
$60'$	8.49	excluded (tie; patched 13.66)	excluded (patched 21.22)	honest (2.445)
$300'$	6.17	patched (3.430)	patched (5.592)	honest (0.686)
$600'$	5.17	patched (1.612; honest 2.694)	patched (2.6933; honest 2.6939)	honest (0.343)
$1800'$	3.58	patched (0.483; honest 0.623)	honest (0.623; patched 0.843)	honest (0.114)

At the exact ties the constants decide ($\|\mathrm{Dom}\| = 4$ against $k_\star + h_\star = 80$). In code R the honest treatment overtakes the restriction after about $\lambda \cdot 76/(\gamma - m^\star_{\mathrm{hon}})$ of the Moon's data in expectation, about 12 at $\lambda = 1$ and 25 at $\lambda = 2$ at $\Delta = 10'$ and $60'$.

For retention, let a datum proved from T cost in addition $r_{\mathrm{d}}\,\mathrm{dl}_T$ bits, where dl_T is the size of the derivation used, and let $J^{\mathrm{der}}_\lambda(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T) + \sum_{\text{data proved}} r_{\mathrm{d}}\,\mathrm{dl}_T$. Compare an honest old law T_{old} (tolerance $\varepsilon_{\mathrm{old}}$ on a window) with an exact successor T_{new} that proves every datum, with per-datum derivation charges $c_{\mathrm{old}} < c_{\mathrm{new}}$.

Proposition A.36 (Retention of a superseded theory)**.**

(a) T_{old} *beats* T_{new} *on* D *iff*

$$\lambda(\|T_{\mathrm{new}}\| - \|T_{\mathrm{old}}\|) + (c_{\mathrm{new}} - c_{\mathrm{old}})\,N_{\mathrm{p}} + c_{\mathrm{new}}\,(n - N_{\mathrm{p}}) > \mathrm{Exc}_D(T_{\mathrm{old}}),$$

where N_{p} *is the number of data that* T_{old} *proves and* $n = |D|$*.*

(b) With i.i.d. data the rates are $m_{\mathrm{old}} = s_{\mathrm{old}}\gamma_{\mathrm{si}} + (1 - s_{\mathrm{old}})c_{\mathrm{old}}$*, with* s_{old} *the silence probability of* T_{old}*, and* $m_{\mathrm{new}} = c_{\mathrm{new}}$*. If* $m_{\mathrm{old}} < m_{\mathrm{new}}$ *(respectively* $>$*), then almost surely* T_{old} *(respectively* T_{new}*) is eventually the* λ*-MAP of the two, whatever* λ *and the lengths are. At* $m_{\mathrm{old}} = m_{\mathrm{new}}$ *the lengths decide if the cost difference is almost surely constant, and otherwise there is no eventual winner.*

(c) Let $s_{\mathrm{old}} = \min(2\varepsilon_{\mathrm{old}}/\Delta, 1)$*. Under code Q,* m_{old} *is non-increasing in* Δ *on* $\{\Delta : \gamma(\Delta) \geq c_{\mathrm{old}}\}$*; under code R it is non-increasing in* Δ *on* $(0, \infty)$ *if* $\beta_e \geq c_{\mathrm{old}}$*. In either case the cell widths in that range at which* T_{old} *is retained form an interval with no upper bound inside the range; call its lower end* Δ^**. For* $\Delta \geq 2\varepsilon_{\mathrm{old}}$ *under code Q,* Δ^* *solves* $2\varepsilon_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})/\Delta = c_{\mathrm{new}} - c_{\mathrm{old}}$*.*

Proof. (a) $J^{\mathrm{der}}_\lambda(T_{\mathrm{new}}) = \lambda\|T_{\mathrm{new}}\| + c_{\mathrm{new}}n$ and $J^{\mathrm{der}}_\lambda(T_{\mathrm{old}}) = \lambda\|T_{\mathrm{old}}\| + \mathrm{Exc}_D(T_{\mathrm{old}}) + c_{\mathrm{old}}N_{\mathrm{p}}$; compare term by term. (b) T_{old} is honest, so it never refutes a datum; a datum costs c_{old} when it proves it and γ_{si} when it is silent. Apply Theorem A.20. (c) *Code Q.* In the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})$, and s_{old} and $\gamma(\Delta) - c_{\mathrm{old}} \geq 0$ are both non-negative and non-increasing in Δ, so their product is too; in the fine regime $m_{\mathrm{old}} = \gamma(\Delta)$, which is decreasing. *Code R.* In the fine regime T_{old} is silent on every datum and $m_{\mathrm{old}} = \mathbb{E}\bar{c}_\Delta(w)$, strictly decreasing by Lemma A.18(iii); in the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\beta_e - c_{\mathrm{old}})$, non-increasing when $\beta_e \geq c_{\mathrm{old}}$. The two forms agree at $\Delta = 2\varepsilon_{\mathrm{old}}$. A non-increasing rate stays below c_{new} once it is below it, which gives the interval, and the equation for Δ^* is $m_{\mathrm{old}} = c_{\mathrm{new}}$ in the coarse regime under code Q. □

With derivations from the successor assumed a thousand times larger ($\mathrm{dl}_{T_{\mathrm{new}}}/\mathrm{dl}_{T_{\mathrm{old}}} = 1000$, as for a validated integration against solving Kepler's equation), exact code-R costs and bisection (code in the accompanying repository) give the least retained cell width Δ^* for $c_{\mathrm{new}} = 0.01$, 0.1 and 1 bit per datum:

body, window	ε^*	code Q	code R
Mercury, 100 yr	$2.02'$	$1542'$, $258.5'$, $37.2'$	$229.8'$, $22.97'$, $2.33'$
Mars, 12 yr	$1.44'$	$1202'$, $195.6'$, $27.7'$	$174.2'$, $17.42'$, $1.744'$
Mars, 2000 yr	$49.9'$	$10450'$, $2895'$, $533'$	$5717'$, $572'$, $57.8'$
Jupiter, 100 yr	$6.57'$	$3468'$, $661'$, $101.6'$	$867'$, $86.7'$, $8.61'$
Saturn, 2000 yr	$81.9'$	never, $3994'$, $784'$	$10320'$, $1032'$, $102.9'$
Moon, 1 month	$30.9'$	$8414'$, $2086'$, $364'$	$4473'$, $447'$, $44.4'$

Under random completion the retention region reaches precisions 1.8 to 6.9 times finer than under literal pricing at $c_{\text{new}} = 0.01$, 3.9 to 11 times at 0.1, and 7.6 to 16 times at 1, because the honest law's silence is cheap.

Question A.37. What is the posterior over an explicit pool of domain predicates (base predicates such as "planet", windows, and their conjunctions, with stated syntax costs) on the Kepler residuals? By Proposition A.35 we expect the answer to depend on the pricing: tolerances with error models under SAI's pricing, and sharp domains under literal pricing at fine precision.

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